Generic standard to demonstrate the compliance of electronic and electrical apparatus with the basic restrictions related to human exposure to electromagnetic fields (0 Hz - 300 GHz)

The European Standard EN 50392:2004: has the status of a British Standard

ICS 13.280

British Standards

National foreword

This British Standard is the official English language version of EN 50392:2004.

The UK participation in its preparation was entrusted to Technical Committee GEL/106, Human exposure to Lf and Hf electromagnetic radiation, which has the responsibility to:

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— present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;

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EUROPEAN STANDARD **EN 50392**

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ICS 13.280

English version

Generic standard to demonstrate the compliance of electronic and electrical apparatus with the basic restrictions related to human exposure to electromagnetic fields (0 Hz - 300 GHz)

Norme de base pour démontrer la conformité des appareils électriques et électroniques, aux restrictions de base pour l'exposition du corps humain aux champs électromagnétiques (0 Hz - 300 GHz)

Fachgrundnorm zur Demonstration der Konformität elektronischer und elektrischer Geräte mit den Basisgrenzwerten für die Exposition von Personen gegenüber elektromagnetischen Feldern (0 Hz - 300 GHz)

This European Standard was approved by CENELEC on 2003-09-23. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CENELEC member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CENELEC member into its own language and notified to the Central Secretariat has the same status as the official versions.

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CENELEC

European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

Central Secretariat: rue de Stassart 35, B - 1050 Brussels

Ref. No. EN 50392:2004 E

Foreword

This European Standard was prepared by the Technical Committee CENELEC TC 106X, Electromagnetic fields in the human environment.

The text of the draft was submitted to the Unique Acceptance Procedure and was approved by CENELEC as EN 50392 on 2003-09-23.

The following dates were fixed:

- latest date by which the EN has to be implemented at national level by publication of an identical national standard or by endorsement	(dop)	2004-10-01
- latest date by which the national standards conflicting with the EN have to be withdrawn	(dow)	2006-10-01

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Contents

1 Scope

The scope of this standard is limited to apparatus which is intended for use by the general public as defined in the Council Recommendation 1999/519/EC of 12 July 1999 on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz) (Official Journal L 199 of 30 July 1999).

This generic standard applies to electronic and electrical apparatus for which no dedicated product- or product family standard regarding human exposure to electromagnetic fields applies.

This generic standard does not cover equipment, which fulfils the requirements given in EN 50371 or is medical equipment as defined in the Council Directive 93/42/EEC of 14 June 1993 concerning medical devices.

The frequency range covered is 0 Hz to 300 GHz.

The object of this standard is to demonstrate the compliance of such apparatus with the basic restrictions or reference levels on exposure of the general public related to electric, magnetic, electromagnetic fields and induced and contact current.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 50357 Evaluation of human exposure to electromagnetic fields from devices used in Electronic Article Surveillance (EAS), Radio Frequency Identification (RFID) and similar applications

EN 50366 Household and similar electrical appliances – Electromagnetic fields – Methods for evaluation and measurement

EN 50371 Generic standard to demonstrate the compliance of low power electronic and electrical apparatus with the basic restrictions related to human exposure to electromagnetic fields (10 MHz - 300 GHz) - General public

EN 50383 Basic standard for the calculation and measurement of electromagnetic field strength and SAR related to human exposure from radio base stations and fixed terminal stations for wireless telecommunication systems (110 MHz - 40 GHz)

EN ISO/IEC 17025:2000 General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:1999)

Council Recommendation 1999/519/EC of 12 July 1999 on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz) (Official Journal L 199 of 30 July 1999)

International Commission on Non-Ionizing Radiation Protection (1998), Guidelines for limiting exposure in time-varying electric, magnetic, and electromagnetic fields (up to 300 GHz). Health Physics 74, 494-522

3 Terms and definitions

For the purposes of this European Standard the following terms and definitions apply.

3.1
root-mean-square (rms)
effective value or the value associated with joule heating, of a periodic electromagnetic wave. The rms value is obtained by taking the square root of the mean of the squared value of a function

NOTE Although many survey instruments indicate rms, the actual quantity measured is root-sum-square (rss) (equivalent field strength). The value rss is obtained from three individual rms field strength values, measured in three orthogonal directions combined disregarding the phases. The measured rss value is the maximum possible (worse case) and can be quite different from the true root-mean-square (rms) value.

3.2
root-sum-square (rss)
effective value or the value associated with joule heating, of a periodic electromagnetic wave. The rss value is obtained by taking the square root of the sum of the squared value of a function

$$X = \sqrt{\sum_{1}^{n} (X_n)^2}$$

3.3
electric field strength (E)
magnitude of a field vector at a point that represents the force (F) on an infinitely small charge (q) divided by the charge

$$E = \frac{F}{q}$$

3.4
magnetic flux density (B)
magnitude of a field vector that is equal to the magnetic field H multiplied by the permeability (μ) of the medium

$$B = \mu H$$

3.5
magnetic field strength (H)
magnitude of a field vector in a point that results in a force (F) on a charge (q) moving with velocity (v)

$$F = q(v \times \mu H)$$

[or magnetic flux density divided by permeability of the medium, see “magnetic flux density”]

3.6
averaging time (t_{avg})
appropriate time over which exposure is averaged for purposes of determining compliance

3.7
basic restriction
basic restriction is the ceiling level that should not be exceeded under any conditions

3.8
contact current
current flowing into the body resulting from contact with a conductive object in an electromagnetic field. This is the localised current flow into the body (usually the hand, for a light brushing contact)

3.9
current density (J)
current per unit cross-sectional area flowing area inside the human body as a result of direct exposure to electromagnetic fields

3.10
duty factor (or duty cycle)
ratio of pulse duration to the pulse period of a periodic pulse train. Also, maybe a measure of the temporal transmission characteristic of an intermittently transmitting RF source such as a paging antenna by dividing average transmission duration by the average period for transmissions. A duty factor of 1.0 corresponds to continuous operation

3.11
exposure
exposure occurs whenever and wherever a person is subjected to electric, magnetic or electromagnetic fields or to contact current other than those originating from physiological processes in the body and other natural phenomena

3.12
exposure level
value of the quantity used to assess exposure. This may be an induced current density, SAR, power density, electric or magnetic field strength, a limb current or a contact current

3.13
exposure, direct effect of
result of a direct interaction in the exposed human body from exposure to electromagnetic fields

3.14
exposure, indirect effect of
result of a secondary interaction between the exposed human body and an electromagnetic field, often used to describe a contact current, shock or burn arising from contact with a conductive object

3.15
exposure, partial-body
localised exposure of part of the body, producing a corresponding localised SAR or induced current density, as distinct from a whole-body exposure

3.16
exposure, whole-body
exposure of the whole body (or the torso when induced current density is considered

3.17
limb current
current flowing in an arm or a leg, either as a result of a contact current or else induced by an external field

3.18
induced current
current induced inside the body as a result of direct exposure to electromagnetic fields

3.19
multiple frequency fields
superposition of two or more electromagnetic fields of differing frequency. These may be from different sources within a device e.g. the magnetron and the transformer of a microwave oven, or they may be harmonics in the field of a nominally single frequency source such as a transformer

3.20
power density (S)
power per unit area normal to the direction of electromagnetic wave propagation. For plane waves the power density (S), electric field strength (E) and magnetic field strength (H) are related by the impedance of free space, i.e. 377 Ω.[1)]

$$S = \frac{E^2}{377} = 377\ H^2 = EH$$

In particular where E and H are expressed in units of W/m² and A/m, respectively, and S in the unit of W/m².

It should be noted that the value of 377 Ω is only valid for free space, far field measurement conditions

3.21
power density, average (temporal)
instantaneous power density integrated over a source repetition period. This averaging is not to be confused with the measurement averaging time

3.22
power density, plane-wave equivalent
commonly used term associated with any electromagnetic wave, equal in magnitude to the power density of a plane wave having the same electric (E) or magnetic (H) field strength as the measured field

3.23
reference levels

reference levels are levels of field strength that can be compared with measured exposure field strengths. The reference levels are derived from the basic restrictions using worst-case assumptions about exposure. If the reference levels are met, then the basic restrictions will be complied with, but if the reference levels are exceeded, that does not necessarily mean that the basic restrictions will not be met

3.24
specific absorption rate (SAR)
power absorbed by (dissipated in) an incremental mass contained in a volume element of biological tissue when exposure to an electromagnetic field occurs. SAR is expressed in the unit watt per kilogram (W/kg). SAR is used as a measure of whole-body exposure as well as localised exposure

3.25
specific absorption (SA)
energy absorbed per unit mass of biological tissue, (SA) expressed in joule per kilogram (J/kg); specific energy absorption is the time integral of specific energy absorption rate

———————

1) Although many survey instruments indicate power density units, the actual quantities measured are E or H or the square of those quantities.

3.26
apparatus
see definitions in the relevant EEC Directives

3.27
EUT
equipment under test (EUT)

3.28
exposure limits
exposure limits are the basic restrictions as specified in Annex II of Council Recommendation 1999/519/EC

4 Compliance criteria

The electronic and electrotechnical apparatus shall comply with the basic restriction as specified in Annex II of Council Recommendation 1999/519/EC.

NOTE 1 The time averaging in the EU-Recommendation applies.

The reference levels in the Council Recommendation 1999/519/EC on public exposure to electromagnetic fields are derived from the basic restrictions using worst-case assumptions about exposure. If the reference levels are met, then the basic restrictions will be complied with, but if the reference levels are exceeded, that does not necessarily mean that the basic restrictions will not be met. In some situations, it will be necessary to show compliance with the basic restrictions directly, but it may also be possible to derive compliance criteria that allow a simple measurement or calculation to demonstrate compliance with the basic restriction. Often these compliance criteria can be derived using realistic assumptions about conditions under which exposures from a device may occur, rather than the conservative assumptions that underlie the reference levels.

NOTE 2 The limit is the basic restriction.

If the technology in the apparatus is not capable of producing an E-field, H-field or contact current, at the normal user position, at levels higher than 1/2 the limit values then the apparatus is deemed to comply with the requirements in this standard in respect of that E-field, H-field or contact current without further assessment.

5 Assessment methods

One or more of the examples of assessment methods in 7.2 may be used.

The assessments should be made according to an existing basic standard. If the assessment method in the basic standard is not fully applicable then deviations are allowed as long as

- a description of the used assessment method is given in the assessment report,
- an evaluation of the total uncertainty is given in the assessment report.

NOTE The information in the annexes is for information and will be replaced by reference to basic standards as soon as they are available.

For transmitters intended for use with external antennas at least one typical combination of transmitter and antenna shall be assessed. The technical specification of this antenna shall be documented in such details that the boundary where the basic restrictions are met can be identified e.g. by documented radiation patterns.

6 Evaluation of compliance to limits

The equipment is deemed to fulfil the requirements of this standard if the measured values are less than or equal to the limit and the assessment uncertainty is less than the measurement uncertainty specified for the applied assessment method(s).

If the assessment uncertainty * is larger than the specified uncertainty value (%) the difference shall be added to the assessment result before comparison with the limit.

NOTE Equation 1 is valid if the measurement uncertainty of the applicable assessment method is 30 % or more.

* The formula for this condition is given below:

$$L_m \leq \left(\frac{1}{0,7 + \frac{U(L_m)}{L_m}} \right) L_{lim} \tag{1}$$

where

L_m is the measured value;

L_{lim} is the exposure limit;

$U(L_m)$ is the expanded uncertainty.

EXAMPLE

Suppose the relative uncertainty of a certain EMF test is 55 %. Then

$$\frac{U(L_m)}{L_m} = 0,55 \; .$$

Using equation (1), the condition for the measured value is then:

$$L_m \leq \left(\frac{1}{0,7 + \frac{U(L_m)}{L_m}} \right) L_{lim} = \left(\frac{1}{0,7 + 0,55} \right) L_{lim} = \frac{1}{1,25} L_{lim} = 0,8 \, L_{lim}$$

The uncertainty penalty is then:

$$U_{pen} \leq 0,2 \, L_{lim}$$

The uncertainty values specified under each assessment method is the maximum allowed uncertainty. If the uncertainty value is not specified then a default value of 30 % shall be used.

7 Applicability of compliance assessment methods

An analysis can be made to investigate what parts emit EMF. A description of the several parts of an apparatus is recommended in order to determine what parts emit EMF. What kind of emission is to be expected under normal conditions.

7.1 Characteristics and parameters of apparatus to be considered

	Further detailed description of the information needed
Frequency	Frequency of emissions
Waveform	Waveform and other information such as duty factor for establishment of peak- and/or average emission
Multiple frequency sources	Does the device produce fields at more than one frequency or fields with a high harmonic content? Are the emissions simultaneous?
Emission of electric fields	Voltage differences and any coupling parts e.g. metallic surfaces charged at a voltage potential
Emission of magnetic fields	Current flow and any coupling parts e.g. coils, transducers or loops
Emission of electromagnetic fields	Generation or transmission of high frequency signals and any radiating parts e.g. antennas, loops, transducers and external cables
Contact currents	Possibility of touching conducting surfaces when either the surface or the person is exposed to electromagnetic fields?
Whole body exposure	Fields produced by device extend over region occupied by the whole body
Partial body exposure	Fields produced by device extend over only part of region occupied by a part of the body, or over region occupied by limbs
Duration/time variation	Duty cycle of emissions, on/off time of power used or emitted by device. Variation of power use or emissions during production process
Homogeneity	Extent to which the strength of the fields varies over the body or region of the body that is exposed. * Shall be measured without the presence of a body
Far – near field	Are exposures in inductive or reactive near field? Propagating near field? Far field?
Pulsed/transient fields	Are the emissions pulse-modulated or true pulses? Are there occasional or periodic transients in the field?
Physical size	Is the device so small that any significant exposure will be to part of the body? In relation to the wavelength (operating frequency) Is it so big that different parts will contribute to exposures “independently”?
Power	What is the emitted power? What is the power consumption? If there is an antenna system, what is the EIRP?
Distance (source ⇒ user)	What is the spatial relationship between the device and the operator or user when it is used normally? The distance used for the assessment shall be specified by the manufacturer and be consistent with the intended usage of the apparatus

Intended usage	How is the device commonly used? Conditions of intended usage producing the highest emission or absorption Operating conditions? How does the intended usage affect the spatial relationship between the device and the user? Can the usage affect the emission characteristics of the device? Can the apparatus be part of a system?
Interaction sources/user	Do the emitted fields change if the device is close to the body? Does the device couple to the body during use?

7.2 List of possible assessment methods

	Applicability area and limitations	**Reference**
Far field calculation	Electromagnetic fields far from source. Very small microwave devices not used close to body, or large lower-frequency transmitters at greater distances. That region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna. In this region (also called the free space region), the field has a predominantly plane-wave character, i.e. locally uniform distribution of electric field strength and magnetic field strength in planes transverse to the direction of propagation.	See Annex A
Near field calculation	Electromagnetic fields very close to the source. There can be an interaction between the radiated fields from the source and the user	See Annex A
Simulation with/without a phantom	Evaluation of measurement results inside the phantom representing a body.	See Annex B
Numerical modelling	Calculation only.	See Annex C
Measurement of body current	Measurement only.	See Annex D
SAR	Calculation and measurements; 100 kHz – 10 GHz.	See Annex E
E&H measurement	Near or far field. Direct measurement for comparison with reference levels or as input for more detailed assessment.	See Annex F
Source modelling	Prediction of exposures from calculation of emissions at a specific distance.	See Annex G
Direct measurement of physical properties: Contact current Input power Body current		See Annex D, E or F

NOTE The physical characteristics and intended use of the apparatus may have an impact on the choice of assessment method. E.g. radiators of EMF intended for use in close proximity to the body shall be assessed differently from transmitters intended for fixed installation in buildings.

7.3 Decision tree (Assessment applicability table versus equipment characteristics)

7.3.1 Generic procedure for assessment of apparatus

1) The apparatus should be characterised to determine the nature of EMF emissions (see 7.2) and also the intended usage conditions.

2) If the device fulfils the requirements of EN 50371 (low power, above 10 MHz) it should not be tested against this standard. If it is inherently incapable of producing any exposure greater then 1/2 the limit values then it is deemed to meet the requirements in this standard in respect of that exposure quantity (Clause 3). If an exposure assessment is necessary then:

3) Fields and body currents should be determined at the typical user position under normal operating conditions by measurement or calculation (see 7.2). If these quantities are below the relevant reference levels, taking into account waveform/frequency content (Clause 7), and any allowed time and spatial averaging then the apparatus is deemed to meet the requirements in this standard. If not, then:

4) Measured exposure levels should be compared with any product-specific compliance criteria that can be derived for the apparatus (Clause 3). If the exposure levels are below the product-specific compliance criteria then the apparatus is deemed to meet the requirements in this standard. If no product-specific compliance criteria have been specified for an E-field, H-field or contact current which is to be assessed, or if compliance criteria have been specified but not met, then:

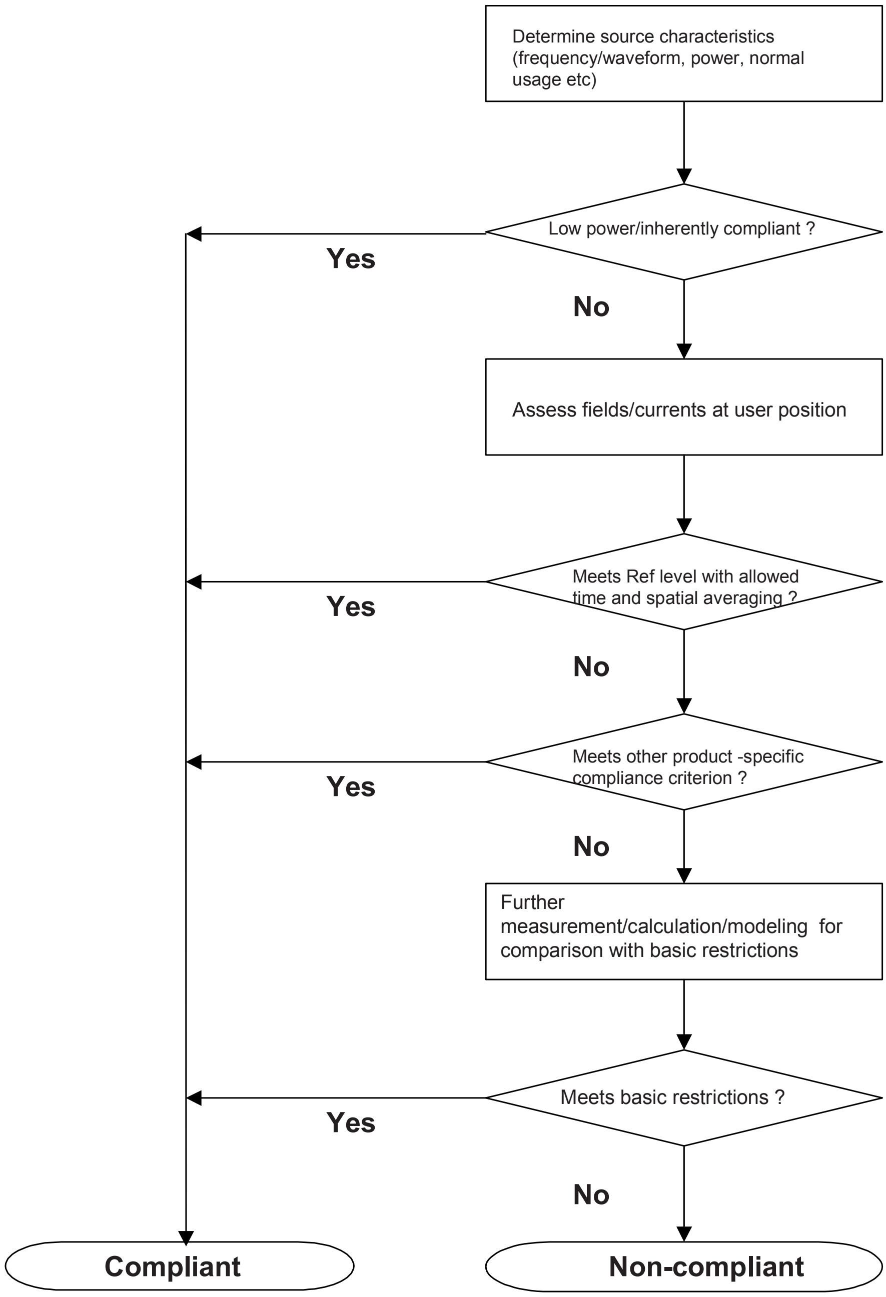
   NOTE The technology of some products may allow assumptions about human exposure from the device to be made e.g. always magnetic field, always partial body etc. From these assumptions it may be possible to derive compliance criteria for that product or product type, e.g. "if the magnetic field strength is below." or "if the power is below.

5) Further assessment involving more detailed measurement, calculation and source/exposure modelling should be undertaken (see 7.2) to allow comparison of exposure levels with all relevant basic restrictions on exposure. If the exposures are below the basic restrictions then the apparatus is deemed to meet the requirements in this standard. If not, then the apparatus is deemed not to comply with the requirements in this standard.

This process is summarised in the flowchart below.

The choice of assessment method in stages 3) and 5) above is optional, but it must be suitable for the exposure quantity to be assessed and for the frequency of emission. Where more than one equally valid assessment method exists for a particular exposure quantity, then it is acceptable to use only one assessment method for that particular quantity. Where only one assessment method is chosen this should be clearly stated and the reasons given for the choice.

Examples of alternative compliance criteria related to e.g. output power and magnetic field characteristics can be found in EN 50357 and EN 50371.



8 Sources with multiple frequencies

In situations where simultaneous exposure to fields of different frequencies occurs phase information is relevant. The effects of these exposures depend on the phase correlation (phase coherent source) between the occurring frequencies. Calculations taking these phase correlations into account are presented in 8.1.

If the sources are independent (phase non-coherent source) the possibility that these exposures will be additive in their effects must be considered. Calculations based on such additivity are shown in 8.2 and should be performed separately for each effect; thus separate evaluations should be made for thermal and electrical stimulation effects on the body.

8.1 Harmonics, pulses and non-sinusoidal fields

8.1.1 Introduction

There are two separate summation regimes for simultaneous exposure to fields of different frequencies: 1 Hz - 10 MHz and 100 kHz - 300 GHz. Exposures to fields with frequencies below 100 kHz should never be added to exposures to fields with frequencies above 10 MHz.

8.1.2 Frequency range from 1 Hz - 10 MHz

In this frequency range the underlying basic restriction is induced current density. Multiple current densities at different frequencies should be undertaken according to the following formula:

$$\sum_{i=1\,\mathrm{Hz}}^{10\,\mathrm{MHz}} \frac{J_i}{J_{L,i}} \leq 1$$

where

J_i is the current density at frequency *i*;

$J_{L,i}$ is the current density basic restriction at frequency *i* as given in Table 1 of the EC Recommendation.

When electric and magnetic field strengths are measured, the exposures should be summed according to these formulae:

$$\sum_{i=1\,\mathrm{Hz}}^{150\,\mathrm{kHz}} \frac{E_i}{E_{L,i}} + \sum_{i>150\,\mathrm{kHz}}^{10\,\mathrm{MHz}} \frac{E_i}{a} \leq 1$$

and

$$\sum_{j=1\,\mathrm{Hz}}^{150\,\mathrm{kHz}} \frac{H_j}{H_{L,j}} + \sum_{j>150\,\mathrm{kHz}}^{10\,\mathrm{MHz}} \frac{H_j}{b} \leq 1$$

where

E_i is the electric field strength at frequency *i*;

$E_{L,i}$ is the electric field strength reference level at frequency *i*;

H_j is the magnetic field strength at frequency *j*;

$H_{L,j}$ is the magnetic field strength reference level at frequency *j*;

a is 87 V/m;

b is 5 A/m (6,25 μT).

For contact current, the following requirements should be applied:

$$\sum_{n=1\,\text{Hz}}^{110\,\text{MHz}} \frac{I_n}{I_{C,n}} \leq 1$$

where

I_n is the contact current component at frequency n;

$I_{C,n}$ is the reference level for contact current at frequency n.

The induced current-density-based summations may or may not include consideration of phase. The most conservative is to neglect phase altogether.

If the frequency components have no constant phase relationship, or the relative phases are not measured (for example if a spectrum analyser is used), an rss summation of frequency components can be undertaken. This will usually give a more realistic outcome than neglecting phase completely, but may still be problematic for pulses and non-sinusoidal fields with harmonics in constant phase relationship: under some very specific circumstances (e.g. Dirac delta pulse) it may be insufficiently protective.

Most realistic is to include relative phase. This can be achieved using either a waveform capture approach, with *post hoc* Fourier analysis [example in EN 50366], or a physical measurement system, which incorporates a "weighting circuit". For comparison with the European Recommendation exposure levels, the weighting circuit should have a frequency response (transfer function A), which matches the frequency response of the exposure standard (function V) so that the weighting and summation of spectral components happens in the time-domain [example in EN 50366].

NOTE Further guidance on the restriction of weighted field values can be found in the ICNIRP statement "Guidance on determining compliance of exposure to pulsed and complex non-sinusoidal waveforms below 100 kHz with ICNIRP guidelines" (Health Physics, vol 84, No 3, pp. 383-387, 2003). This approach is based on the restriction of the weighted peak value of a broadband field. The weighting function has been derived from the reference levels as a function of frequency. The weighted peak restriction can be applied for periodic non-sinusoidal waveforms where the mutual phases of harmonic components do not vary significantly.

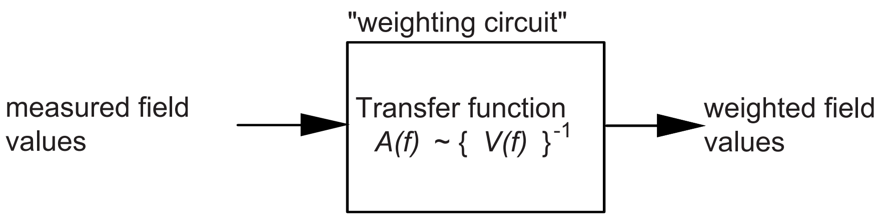


Figure 1 – Schematic of "weighting circuit"

EXAMPLE: weighting function used in EN 50366

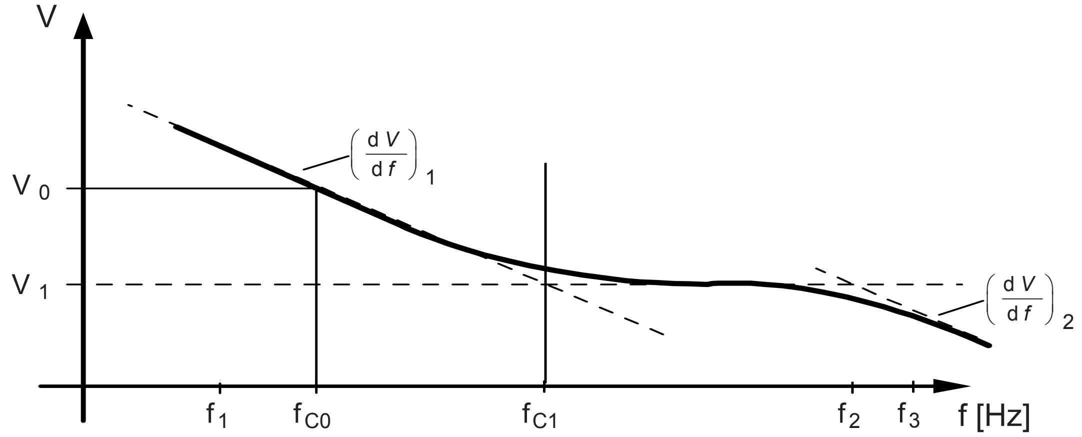


Figure 2 – Dependency on frequency of the reference levels V with smoothed edges

with $V(f_{C0}) = V_0$, $V(f_{C1}) = V_1$ and the gradients $\left(\frac{dV}{df}\right)_n$

The transfer function A is the on V_0 normalized inverse of the reference level V. The normalization shall be done at the frequency f_{C0}.

The transfer function A shall have the following characteristics (shown in double logarithmic scale) and shall be realized with a first order filter:

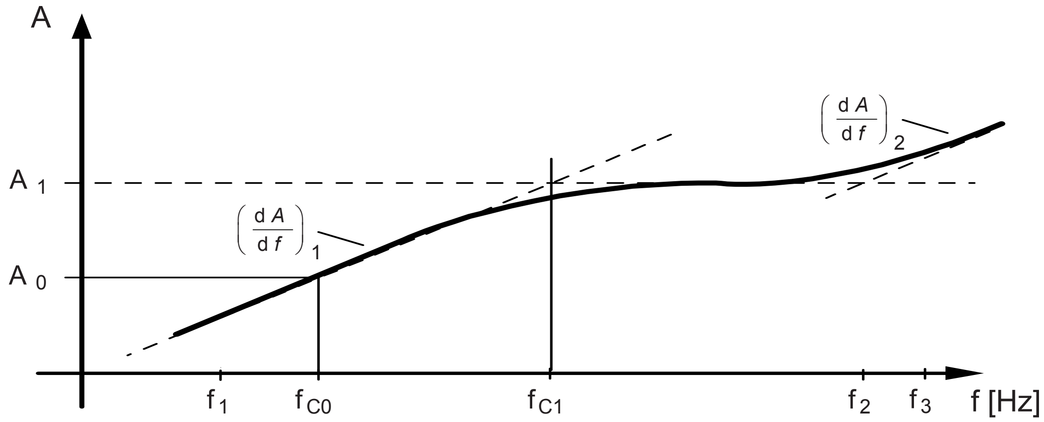


Figure 3 – Transfer function A

$$A(f) = \frac{V(f_{C0})}{V(f)}$$

For the transfer function the following shall be suitable:

$$A(f_{C0}) = A_0 = \frac{V(f_{C0})}{V_0} = 1, \qquad A(f_{C1}) = A_1 = \frac{V(f_{C0})}{V_1},$$

and for the gradients $\left(\frac{dA}{df}\right)_n = \left[\left(\frac{dV}{df}\right)_n\right]^{-1}$

The reference level $B_{RL}(f)$ of the EC Recommendation can be used to calculate the transfer function as follows:

$V(f) := B_{RL}(f)$

$(f_1 = 10\text{ Hz}) \le f \le (f_{C1} = 800\text{ Hz})$: $$A(f) = \frac{B_{RL}(f_{C0} = 50\text{ Hz})}{B_{RL}(f)} = \frac{5\,000/50\ \mu\text{T}}{5\,000/f\ \mu\text{Ts}} = \frac{f}{50\text{ Hz}}$$

$(f_{C1} = 800\text{ Hz}) \le f \le (f_2 = 150\text{ kHz})$: $$A(f) = \frac{B_{RL}(f_{C0} = 50\text{ Hz})}{B_{RL}(f)} = \frac{5\,000/50\ \mu\text{T}}{6{,}25\ \mu\text{T}} = 16$$

$(f_2 = 150\text{ kHz}) \le f \le (f_{n=3} = 400\text{ kHz})$: $$A(f) = \frac{B_{RL}(f_{C0} = 50\text{ Hz})}{B_{RL}(f)} = \frac{5\,000/50\ \mu\text{T}}{920\,000/f\ \mu\text{Ts}} = \frac{f}{9{,}2\text{ kHz}}$$

NOTE All frequencies *f* used above in Hz.

8.1.3 Frequency range from 100 kHz - 300 GHz

In this frequency range, the exposure standard is based on the avoidance of thermal effects. *Note words to cover non-thermal.* The basic restrictions are on SAR and power density, and summation of these quantities should follow the formula

$$\sum_{i=100\text{ kHz}}^{10\text{ GHz}} \frac{SAR_i}{SAR_L} + \sum_{i>10\text{ GHz}}^{300\text{ GHz}} \frac{S_i}{S_L} \le 1$$

where

SARs can be for the whole body or part of body. Partial-body SARs should be summed together; whole body SARs should be summed together. Partial body should not be summed with total body.

where

SAR_i is the SAR caused by exposure at frequency *i*;

SAR_L is the SAR basic restriction;

S_i is the power density at frequency *i*;

S_L is the power density basic restriction.

Exposure field strengths can be compared to the reference levels on an rss basis:

$$\sum_{i=100\text{ kHz}}^{1\text{ MHz}} \left(\frac{E_i}{c}\right)^2 + \sum_{i>1\text{ MHz}}^{300\text{ GHz}} \left(\frac{E_i}{E_{L,i}}\right)^2 \le 1$$

and

$$\sum_{i=100\text{ kHz}}^{150\text{ kHz}} \left(\frac{H_i}{d}\right)^2 + \sum_{i>150\text{ kHz}}^{300\text{ GHz}} \left(\frac{H_i}{H_{L,i}}\right)^2 \le 1$$

where

E_i is the electric field strength at frequency *i*;

$E_{L,i}$ is the electric field reference level from Table 2 of the EC Recommendation;

H_i is the magnetic field strength at frequency *i*;

$H_{L,i}$ is the magnetic field reference level derived from Table 2 of the EC Recommendation;

c is $87/f^{½}$ V/m (f in MHz);

d is 0,73/f A/m (f in MHz).

The summation formula for limb current is:

$$\sum_{k=10\,\mathrm{MHz}}^{110\,\mathrm{MHz}} \left(\frac{I_k}{I_{L,k}}\right)^2 \leq 1\,\mathrm{kHz}$$

where

I_k is the limb current component at frequency *k*;

$I_{L,k}$ is the reference level for limb current, 45 mA.

Under this thermal summation regime, the relative phases of the spectral components can be neglected.

9 Assessment report

9.1 General

The results of each assessment, test, calculation or measurement carried out shall be reported accurately, clearly, unambiguously and objectively and in accordance with any specific instructions in the required method(s).

The results shall be recorded, usually in an assessment report, and shall include all the information necessary for the interpretation of the assessment, test or calibration results and all information required by the used method.

All the information needed for performing repeatable assessments, tests, calculations, or measurements shall be recorded.

Further guidelines on the assessment report can be found in EN ISO/IEC 17025, subclause 5.10.

9.2 Items to be recorded in the assessment report

9.2.1 Assessment method

The chosen assessment method shall be recorded including the rationale (see Clause 5) for the choice.

9.2.2 Presentation of the results

- Description of the device / Serial number if applicable
- Testing conditions (temperature, etc.) if applicable
- Operating conditions
- Results of validation check on assessment method
- Results of each performed assessment

9.2.3 Equipment using external antennas

The technical specification of an external antenna shall be documented in such details that the boundary where the basic restrictions are met can be identified e.g. by documented radiation patterns. The characteristics of the transmitter shall also be documented (e.g. output power, frequency, modulation etc.).

10 Information to be supplied with the apparatus

The manufacturer must provide all necessary information with the product with regard to the safe use (also special precautions if needed during repair/maintenance).

Annex A
(informative)

Field calculation

A.1 Purpose

This annex contains the background on "Electromagnetic Field calculation" including the justification of the boundaries between field regions and some support of the formulas used in the calculation methods.

A.2 Far-field region

The field calculation does not take into account the antenna size, which is assumed to be a point source. An ideal isotropic antenna is used as a reference to compare the performance of practical antennas: P watts is radiated, from a point, uniformly over the surface of sphere of radius r.

The POYNTING VECTOR gives the power density: $S = E \wedge H = \dfrac{E^2}{\eta} = \dfrac{P}{4\pi r^2}$

In free space: $E = \eta_0 H = \dfrac{\sqrt{30PG(\theta,\phi)}}{r}$

G = antenna gain relative to an isotropic antenna
θ,ϕ = elevation and azimuth angles to point of investigation
r = distance from observation point to the antenna
η_0 = Characteristic impedance of free space

A.3 Radiating near-field region

Real antennas have finite dimensions (not a point source).

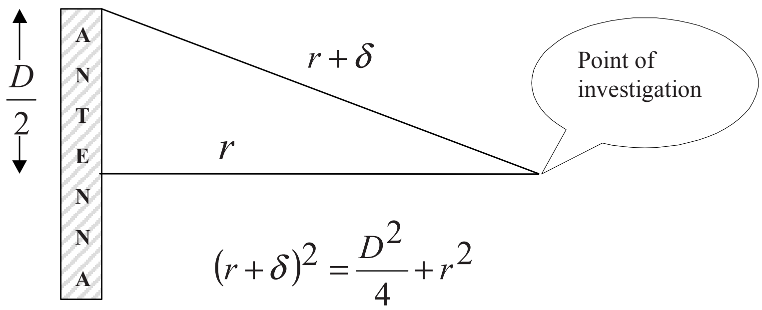


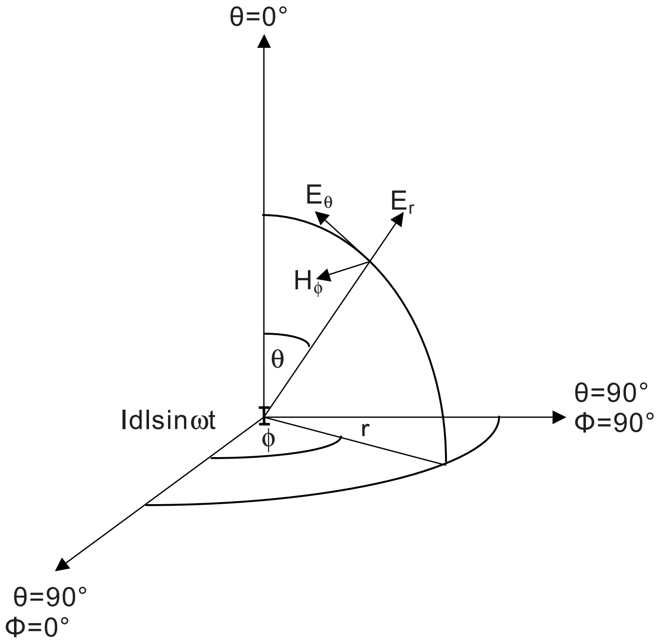
Figure A.1 – Geometry of antenna with largest linear dimension D

The phase difference between the signals from the end of the antenna to that from the centre is a function of the path difference δ. When δ is greater than the Rayleigh criterion of λ/16 this phase difference will significantly modify the signal level at the point of investigation. Thus, when $r \leq \frac{2D^2}{\lambda}$ free space conditions from a point source no longer apply. If r becomes very small the reactive near-field conditions are significant, see Figure A.3 below.

This requires that the boundary for the radiating near-field region be defined by: $\frac{\lambda}{4} < r \leq \frac{2D^2}{\lambda}$ If the antenna is very short, $2D^2/\lambda$ may be less than λ/4. In which case the radiating near field region will be inside the reactive near field region.

A.4 Reactive near field region

Electromagnetic field equations for complicated antenna systems can be derived knowing fields produced by an oscillating current I Sinωt in a short linear element:



$H_\phi = \psi(1 - \alpha)$
$E_\theta = \eta\psi\ (1 - \alpha + \alpha^2)$
$E_r = 2\eta_0\psi\cot(\theta)(\alpha-\alpha^2)$

Where $\Psi = \frac{\mathrm{Idlsin}\theta}{2\lambda\,\mathrm{r}} \mathrm{e}^{-j\omega\left(t-\frac{r}{c}\right)}$ and $\alpha = \frac{j}{\beta r}$

H_ϕ = magnetic field
E_θ= electric field
E_r = radial electric field
I = elemental current, A
dl = element length, m
ω = 2πf, rads/s (f = frequency, Hz)
β = 2π/λ, m^{-1}

Figure A.2 – Current element Idlsin(ωt) at the origin of spherical co-ordinates

α represents induction and α² represents electrostatic near-fields of the reactive near field terms. The energy represented by these terms circulates (ebbs/flows) around the source, i.e. it does not propagate outwards towards infinity.

To determine the difference between the non-radiative and radiative components the following analysis can be performed.

For H² values only, summing the real and imaginary components and dividing by the radiated component we get: $\left\|\frac{\psi(1-\alpha)}{\psi}\right\|^2 = \|1-\alpha\|^2 = \left\|1-j(\frac{\lambda}{2\pi\ r})\right\|^2 = 1+\frac{\lambda^2}{4\pi^2 r^2}$

For E² values only when $\theta \Rightarrow 90°$ (i.e. antenna centre element bore sight) $E_r \Rightarrow 0$, taking real and imaginary components of E_θ dividing by radiated components we get:

$$\left\|\frac{\eta\psi(1-\alpha+\alpha^2)}{\eta\psi}\right\|^2 = \left\|1-\alpha+\alpha^2\right\|^2 = \left\|1-\frac{\lambda^2}{4\pi^2 r^2}-j(\frac{\lambda}{2\pi\ r})\right\|^2 = 1-\frac{\lambda^2}{4\pi^2 r^2}+\frac{\lambda^4}{16\pi^4 r^4}$$

For ExH values

When $\theta \Rightarrow 90°$ (i.e. antenna centre element bore sight) $E_r \Rightarrow 0$, taking real and imaginary components of E_θ and H_ϕ dividing by radiated components we get

$$\left\|\frac{\eta\psi^2(1-\alpha)(1-\alpha+\alpha^2)}{\eta\psi^2}\right\| = \left\|1-2\alpha+2\alpha^2-\alpha^3\right\| = \left\|1-\frac{\lambda^2}{2\pi^2 r^2}-j(\frac{\lambda}{\pi r}+\frac{\lambda^3}{8\pi^3 r^3})\right\| = \sqrt{1+\frac{\lambda^6}{64\pi^6 r^6}}$$

As can be seen, η and ψ terms cancel in the above ratios, thus there are no time or impedance terms present.

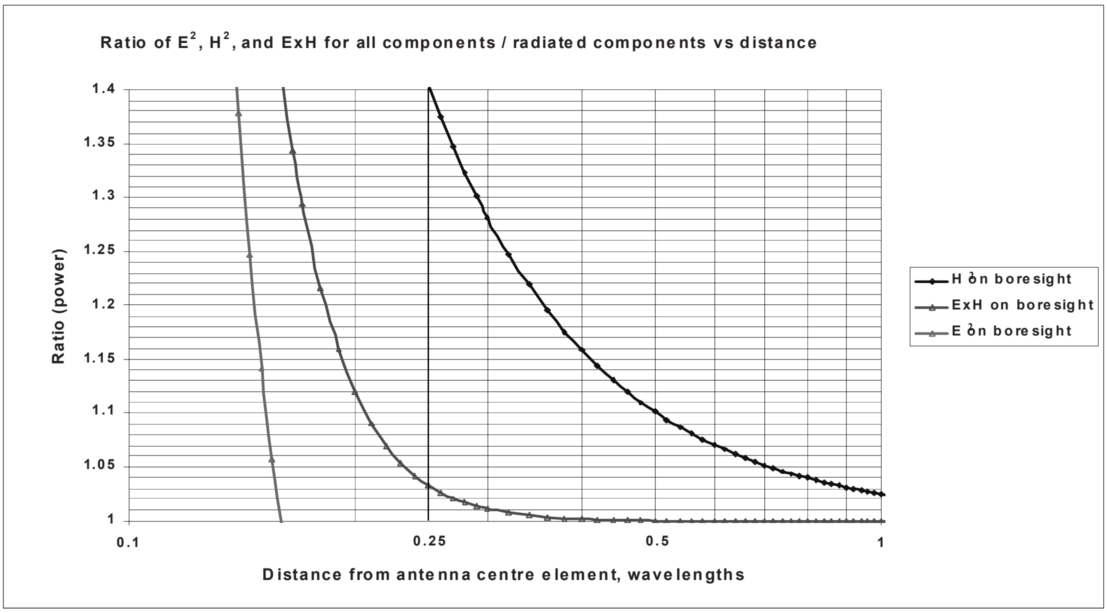


Figure A.3 – Ratio of E², H², and ExH field components

A.4.1 Typical antenna examples

Figure A.4 below shows three example antenna ratios for ExH between all field terms and radiated terms, the graphs were produced using a model based on a vector summation of the infinitesimal wave equations. The 7 dipole array antenna and 12 dipole array antenna were modelled using the approximation of one infinitesimal increment per dipole. The single dipole antenna was divided into 15 equally spaced infinitesimal increments

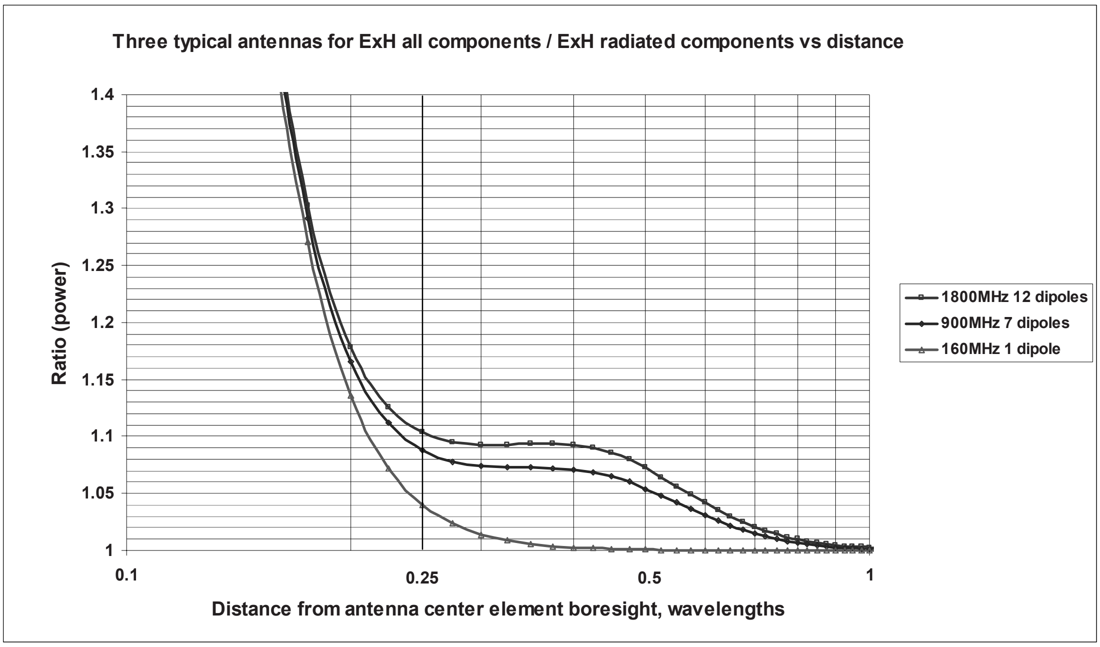


Figure A.4 – Ratio of ExH field components for three typical antennas

A.4.2 Discussion

From Figure A.4 it can be seen the ratio remains at 1,1 or less at distances greater than $\lambda/4$. Thus if a minimum calculation distance of $\lambda/4$ is used, the effective maximum difference between all field components and radiated field components would be 10 % or less, for the three example antennas.

A.4.3 Conclusion

The bore sight ratio of all components divided by radiated components for ExH is $\sqrt{1+\frac{\lambda^6}{64\pi^6 r^6}}$ at close distances to the antenna, giving the result that at a distance of $\lambda/2\pi$ from the antenna the power ratio is 1,41. Unlike the single dipole antenna case, for a multiple dipole antenna, as the distance from the antenna increases from $\lambda/2\pi$ other off centre dipoles contribute to the ratio (the radial E field), but as can be seen from Figure A.4 these increases are marginal.

Recommend $\lambda/4$ as the boundary between the radiated near field and reactive near field for RF exposure compliance assessment.

NOTE This is especially the case when compared with the uncertainty of evaluating SAR.

A.5 Example of calculations within field regions at 900 MHz

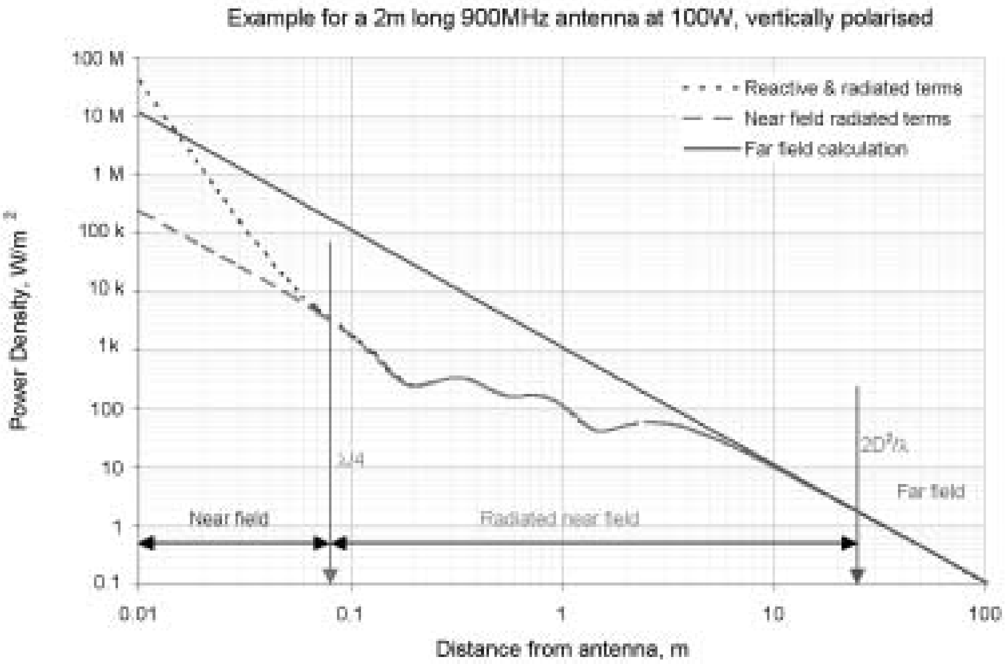


Figure A.5 – Far-field = straight line, radiated near-field = lower line & all near-fields = other line

Annex B
(informative)

SAR compliance assessment

B.1 Whole body SAR

B.1.1 Introduction

The current version of this standard does not include specifications for whole-body SAR measurements. Such measurements are for further study and will be described in later revisions of this standard.

Whole-body SAR measurements are not required for transmitters that have maximum output power levels too low to result in exposure levels that can reach the whole-body SAR compliance limits under any conditions. This section specifies whole-body SAR exclusion criteria.

B.1.2 Whole-body SAR implicit compliance

If the maximum radiated rms power emitted by EUT is less than the values specified in Table B.1, the maximum exposure will not exceed the whole-body averaged SAR compliance limits under any conditions and thus whole-body SAR measurements are not necessary.

Table B.1 - Determining whole-body SAR exclusion power levels

Exposure category	Maximum radiated rms power (W)
Children	Maxpower = $SAR_{Wblimit}$ * 12,5
Adults	Maxpower = $SAR_{Wblimit}$ * 42

Rationale for the whole-body SAR exclusion power levels

The whole-body SAR exclusion power levels have been derived based on the following assumptions:

1) all of the power emitted from the antenna is absorbed in the body (worst-case assumption);
2) the body masses for a 4-year-old child and a 16-year-old worker have been taken as 12,5 kg and 42 kg, respectively. This is the 3rd percentile body weight data for girls and women (conservative approach) (see EN 50383).

B.2 Localised SAR

This clause describes the procedure for measurements of the maximum localised SAR in a phantom model that simulates a person exposed to radio frequency fields emitted by an antenna. The measurement protocol described here shall be used to verify that equipment under test (EUT) is in compliance with the localised SAR limits at a specified distance. It can also be used to determine the compliance distance for a certain output power level or to determine the maximum output power level to meet a compliance distance requirement.

Since the available information about localised SAR measurement methodologies for base station antennas is limited, the procedure described in the current revision of this standard is valid only for the following conditions:

1) the separation between the phantom and the outer surface of the radiating structure shall be 40 cm or less;
2) the size of the radiating structure surface shall be less than 60 cm by 30 cm;
3) the frequency shall be in the range from 30 to 3 000 MHz.

If these conditions are not met, assessments of field strength or power density in air shall be performed.

Since the recommended SAR limits for the limbs are five times higher than for the head and trunk, measurements of SAR in the limbs are not considered. The size of the phantom described in this section has been chosen to correspond to the trunk of an adult man. The phantom is shaped like a box in order to simplify the measurements and the manufacturing of the phantom. The absorption by a box shaped phantom is at least as high as in an anatomically shaped body model.

The same tissue-simulating liquids specified for SAR measurements of handheld mobile phones (see references [1] & [2] under C.9) have also been selected for this standard. The rationale for this is that dielectric parameters of the skin and muscle tissues, which are normally most exposed, are close to those specified for head tissue. This also means that the measurement results are relevant also for head exposure, and that only one set of tissue recipes are needed for SAR testing of mobile or fixed EUTs.

Reference [3] under C.9 indicates that the homogeneous phantom model specified in this standard may give localised SAR values lower than the maximum values in a heterogeneous and anatomically realistic body model. Further studies are needed to verify these results and possibly develop a phantom that provides more accurate estimates of the true maximum localised SAR.

Annex C
(informative)

Information for numerical modelling

C.1 Introduction

This annex provides some information for numerical modelling purposes. Comparisons have been made between different models and methods, with varying degrees of correlation [1,2]. One of the most important areas contributing to the differences is the anatomical body model used and its associated electrical properties.

There are a number of different calculation methods, some of which are mentioned in Clause 5. Comparisons of results have shown good correlation between different numerical calculation methods.

C.2 Anatomical models

During the drafting of this document, a number of anatomical models were identified. References to these, or the institution responsible for them, do not indicate that they are any more suitable or more accurate than other models are. The parameters and voxel size of the model can contribute significant uncertainties, which is why most models are scaled to match the ICRP Standard Man [3].

C.2.1 The Visible Human Project

The Visible Man data set is the first result of the Visible Human Project of the National Library of Medicine, 8600 Rockville Pike, Bethesda, Maryland, USA. It is a digital image data set of a complete human male and consists of computed tomographic and magnetic resonance scans as well as cyrosection images.

C.2.2 “MEET Man”

This is a processed version of the Visible Man data set to obtain a volume data set in voxel representation, which has then been segmented and classified into 40 different tissue types. This work was done by the Institute of Biomedical Engineering, University of Karlsruhe, Kaiserstrasse 12, D-76128 Karlsruhe, Germany.

C.2.3 “Hugo”

This anatomical 3D volume and surface data set is also based on the Visible Man information. The data is currently categorised into 40 types of tissue. The data is created in different forms, including a voxel set, useful for dosimetry. ViewTec, Schaffhauserstrasse 466, CH-8052 Zürich, Switzerland.

C.2.4 “Norman”

This model is a 3D array of voxels, each of which contains information on its discrete tissue type (or air). It is based on medical imaging data and has been categorised into 37 different tissue types and scaled to match the ICRP66 Standard Man. This work was done by the National Radiological Protection Board (NRPB), Chilton, Didcot, Oxfordshire, UK.

C.2.5 University of Utah

This anatomically based voxel model of the human body was obtained from MRI scans of a male volunteer. It is categorised into 31 Tissue types and is scaled to match the ICRP66 Standard Man.

C.2.6 University of Victoria

This is a voxel-based model categorised with up to 128 different tissues. This work has been done by The Applied Electromagnetics Group, Department of Electrical and Computer Engineering, University of Victoria, Victoria, B.C., Canada, V8W 3P6.

C.3 Simpler, homogeneous body models

In order to model the induced current density, a simplified body shape of uniform conductivity can also be used. Suitable body models a prolate spheroids. The dielectric properties of such a model are often the whole body average at the frequencies being investigated, but could, instead, be representative of particular body parts or tissue types which were being investigated. The results are highly dependant on the size of the model and these models tend to overestimate the current density when in the near field.

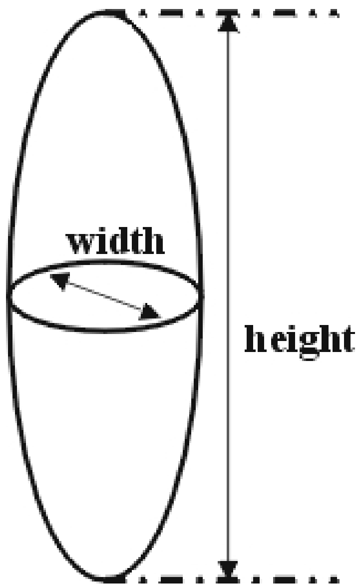


Figure C.1 – Numerical model of an homogenous ellipsoid

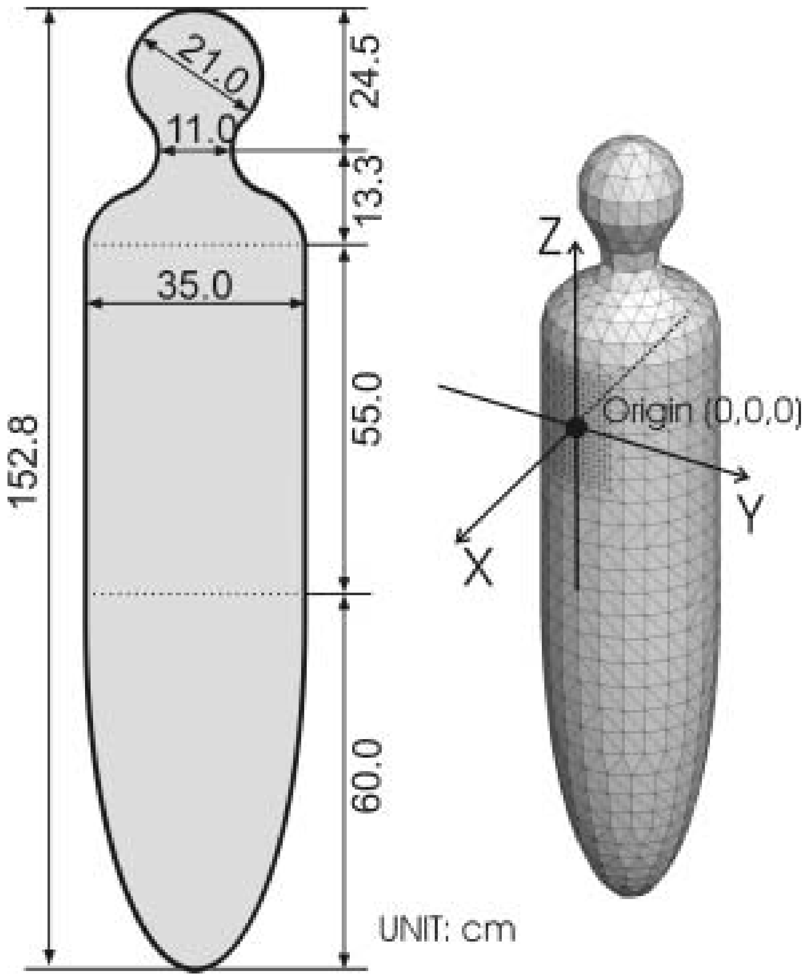


Figure C.2 – Numerical model of an homogenous human body

C.4 Electrical properties of tissue

There have been several investigations into the electrical characteristics of various tissue types [4,5,6,7]. In most cases, they were published for specific frequencies or ranges of frequencies. It has been shown that these properties vary with frequency and values have been interpolated between frequencies and tissue types when modelling. It is also possible that further interpolation and/or averaging of property values is required to match the exact tissue characterisation of particular anatomical models.

Gabriel, et al, made an extensive evaluation of this in published papers and reports during 1995/6. The work included new measurements, a comparison of existing literature and an algorithm to calculate the properties across a wide range of frequencies [8,9,10,11]. This is generally accepted to be the most comprehensive work on the subject, at the date of issue of this Standard. A significant proportion of current modelling work uses these values as a basis, supplementing them with information from previous work where appropriate. The uncertainties grow larger at the ends of the frequency range and this has to be taken into consideration.

Work continues in this field, however, and this may produce new results in the future.

It must be noted that some tissue types are anisotropic (i.e. have different properties in different directions). It is not always possible to model this effect, however, and so an average (or similar) value is used in the model.

The tables of values provided here were obtained from calculations made by the Electromagnetic Wave Research Institute of the Italian National Research Council [11], based on the algorithms provided in the Gabriel report to the Brooks AFB. These tables are example values, which may be used or interpolated for numerical modelling purposes. More precise values, at specific frequencies, may also be obtained from the quoted references or work of a similar nature.

Table C.1 – Conductivity of tissue types

	Conductivity (S/m)									
Frequency	**10 Hz**	**100 Hz**	**1 kHz**	**10 kHz**	**100 kHz**	**1 MHz**	**10 MHz**	**100 MHz**	**1 GHz**	**10 GHz**
Tissue type										
Air	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
Aorta	0,25	0,28	0,31	0,31	0,32	0,33	0,34	0,46	0,73	9,13
Bladder	0,20	0,21	0,21	0,21	0,22	0,24	0,27	0,29	0,40	3,78
Blood	0,70	0,70	0,70	0,70	0,70	0,82	1,10	1,23	1,58	13,13
Bone (Cancellous)	0,08	0,08	0,08	0,08	0,08	0,09	0,12	0,17	0,36	3,86
Bone (Cortical)	0,02	0,02	0,02	0,02	0,02	0,02	0,04	0,06	0,16	2,14
Bone (Marrow)	0,00	0,00	0,00	0,00	0,00	0,00	0,01	0,02	0,04	0,58
Brain (Grey matter)	0,03	0,09	0,10	0,11	0,13	0,16	0,29	0,56	0,99	10,31
Brain (White matter)	0,03	0,06	0,06	0,07	0,08	0,10	0,16	0,32	0,62	7,30
Breast fat	0,02	0,02	0,02	0,02	0,03	0,03	0,03	0,03	0,05	0,74
Cartilage	0,16	0,17	0,17	0,18	0,18	0,23	0,37	0,47	0,83	9,02
Cerebellum	0,05	0,11	0,12	0,13	0,15	0,19	0,38	0,79	1,31	9,77
Cerebro spinal fluid	2,00	2,00	2,00	2,00	2,00	2,00	2,00	2,11	2,46	15,38
Cervix	0,30	0,41	0,52	0,54	0,55	0,56	0,63	0,74	0,99	10,05
Colon	0,01	0,12	0,23	0,24	0,25	0,31	0,49	0,68	1,13	11,49
Cornea	0,41	0,42	0,42	0,44	0,50	0,66	0,87	1,04	1,44	11,33
Duodenum	0,51	0,52	0,52	0,53	0,54	0,58	0,78	0,90	1,23	13,31
Dura	0,50	0,50	0,50	0,50	0,50	0,50	0,54	0,74	0,99	8,58
Eye sclera	0,50	0,50	0,50	0,51	0,52	0,62	0,80	0,90	1,21	11,31
Fat	0,01	0,02	0,02	0,02	0,02	0,03	0,03	0,04	0,05	0,59
Gall bladder	0,90	0,90	0,90	0,90	0,90	0,90	0,90	1,01	1,29	12,53
Gall bladder bile	1,40	1,40	1,40	1,40	1,40	1,40	1,40	1,54	1,88	15,36
Heart	0,05	0,09	0,11	0,15	0,22	0,33	0,50	0,73	1,28	11,84
Kidney	0,05	0,10	0,11	0,14	0,17	0,28	0,51	0,81	1,45	11,57
Lens	0,26	0,26	0,26	0,27	0,28	0,30	0,43	0,56	0,83	8,53
Liver	0,03	0,04	0,04	0,05	0,08	0,19	0,32	0,49	0,90	9,39
Lung (Deflated)	0,20	0,21	0,22	0,24	0,27	0,33	0,44	0,56	0,90	10,12
Lung (Inflated)	0,04	0,07	0,08	0,09	0,11	0,14	0,23	0,31	0,47	4,21
Mucous membrane	0,00	0,00	0,00	0,00	0,07	0,22	0,37	0,52	0,88	8,95
Muscle	0,20	0,27	0,32	0,34	0,36	0,50	0,62	0,71	0,98	10,63
Nerve	0,02	0,03	0,03	0,04	0,08	0,13	0,22	0,34	0,60	6,03
Oesophagus	0,51	0,52	0,52	0,53	0,54	0,58	0,78	0,90	1,23	13,31
Ovary	0,31	0,32	0,32	0,33	0,34	0,36	0,46	0,75	1,34	9,82
Pancreas	0,05	0,10	0,11	0,14	0,17	0,28	0,51	0,81	1,45	11,57
Prostate	0,41	0,42	0,42	0,43	0,44	0,56	0,78	0,91	1,25	12,38

Table C.1 – Conductivity of tissue types *(continued)*

	Conductivity (S/m)									
Frequency	**10 Hz**	**100 Hz**	**1 kHz**	**10 kHz**	**100 kHz**	**1 MHz**	**10 MHz**	**100 MHz**	**1 GHz**	**10 GHz**
Tissue type										
Skin (Dry)	0,00	0,00	0,00	0,00	0,00	0,01	0,20	0,49	0,90	8,01
Skin (Wet)	0,00	0,00	0,00	0,00	0,07	0,22	0,37	0,52	0,88	8,95
Small intestine	0,51	0,52	0,53	0,56	0,59	0,86	1,34	1,66	2,22	12,69
Spinal cord	0,02	0,03	0,03	0,04	0,08	0,13	0,22	0,34	0,60	6,03
Spleen	0,04	0,10	0,10	0,11	0,12	0,18	0,51	0,80	1,32	11,38
Stomach	0,51	0,52	0,52	0,53	0,54	0,58	0,78	0,90	1,23	13,31
Tendon	0,25	0,30	0,38	0,39	0,39	0,39	0,41	0,49	0,76	10,34
Testis	0,41	0,42	0,42	0,43	0,44	0,56	0,78	0,91	1,25	12,38
Thymus	0,51	0,52	0,52	0,53	0,54	0,60	0,72	0,79	1,08	12,13
Thyroid	0,51	0,52	0,52	0,53	0,54	0,60	0,72	0,79	1,08	12,13
Tongue	0,26	0,27	0,27	0,28	0,29	0,39	0,57	0,67	0,98	11,08
Trachea	0,30	0,30	0,30	0,31	0,34	0,37	0,46	0,55	0,80	8,54
Uterus	0,20	0,29	0,49	0,51	0,53	0,56	0,75	0,94	1,31	12,49
Vacuum	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
Vitreous humor	1,50	1,50	1,50	1,50	1,50	1,50	1,50	1,50	1,67	15,13

Table C.2 – Relative permitivity of tissue types

Frequency	100 kHz	1 MHz	10 MHz	100 MHz	1 GHz	10 GHz
Tissue type						
Air	1	1	1,0	1,0	1,0	1,0
Aorta	930	218	109,5	59,8	44,6	32,7
Bladder	1 231	343	51,5	22,7	18,9	14,0
Blood	5 120	3 026	280,0	76,8	61,1	45,1
Bone (Cancellous)	472	249	70,8	27,6	20,6	12,7
Bone (Cortical)	228	145	36,8	15,3	12,4	8,1
Bone (Marrow)	111	40	19,3	6,5	5,5	4,6
Brain (Grey matter)	3 222	860	319,7	80,1	52,3	38,1
Brain (White matter)	2 108	480	175,7	56,8	38,6	28,4
Breast fat	71	24	7,9	5,7	5,4	3,9
Cartilage	2 572	1 391	179,3	55,8	42,3	25,6
Cerebellum	3 515	1 141	464,7	89,8	48,9	34,6
Cerebro spinal fluid	109	109	108,6	88,9	68,4	52,4
Cervix	1 751	448	179,7	60,3	49,6	37,7
Colon	3 722	1 679	271,5	81,8	57,5	41,9
Cornea	10 567	2 878	259,4	76,0	54,8	40,3
Duodenum	2 861	1 678	246,4	77,9	64,8	48,9
Dura	326	253	194,9	60,5	44,2	33,0
Eye sclera	4 745	2 178	208,3	67,9	55,0	41,5
Fat	93	27	13,8	6,1	5,4	4,6
Gall bladder	107	100	98,8	79,0	59,0	47,2
Gall bladder bile	120	120	119,5	95,0	70,0	55,9
Heart	9 846	1 967	293,5	90,8	59,3	42,2
Kidney	7 652	2 251	371,2	98,1	57,9	40,3
Lens	1 704	829	212,5	55,8	41,8	30,7
Liver	7 499	1 536	223,1	69,0	46,4	32,5
Lung (Deflated)	5 145	1 171	180,3	67,1	51,1	38,0
Lung (Inflated)	2 581	733	123,7	31,6	21,8	16,1
Mucous membrane	15 357	1 833	221,8	66,0	45,7	33,5
Muscle	8 089	1 836	170,7	66,0	54,8	42,8
Nerve	5 133	926	155,1	47,3	32,3	23,8
Oesophagus	2 861	1 678	246,4	77,9	64,8	48,9
Ovary	1 942	678	293,6	87,2	49,8	32,8
Pancreas	7 652	2 251	371,2	98,1	57,9	40,3
Prostate	5 717	2 683	246,9	75,6	60,3	45,2
Skin (Dry)	1 119	991	361,7	72,9	40,9	31,3
Skin (Wet)	15 357	1 833	221,8	66,0	45,7	33,5
Small intestine	13 847	5 676	488,5	96,5	58,9	42,0
Spinal cord	5 133	926	155,1	47,3	32,3	23,8
Spleen	4 222	2 290	440,5	90,7	56,6	40,6
Stomach	2 861	1 678	246,4	77,9	64,8	48,9
Tendon	472	160	103,2	53,9	45,6	29,3
Testis	5 717	2 683	246,9	75,6	60,3	45,2
Thymus	3 301	1 433	162,7	68,8	59,5	45,2
Thyroid	3 301	1 433	162,7	68,8	59,5	45,2
Tongue	4 746	2 178	208,3	67,9	55,0	41,5
Trachea	3 735	775	146,1	53,0	41,8	31,1
Uterus	3 411	1 168	321,6	80,0	60,8	45,3
Vacuum	1	1	1,0	1,0	1,0	1,0
Vitreous humor	98	84	70,0	69,1	68,9	57,9

C.5 Calculation of the induced electric current density

Any numerical method and any field calculation software package that is suitable for the models and procedures described in C.8.1 and C.8.2 may be used if passing the given validation procedure. Generally applied methods are:

- BEM (boundary element method);
- FDFD (finite difference frequency domain);
- FDTD (finite difference time domain);
- FEM (finite element method);
- FIT (finite integration technique);
- MoM (method of moments).

If using RF software codes, the application of a **frequency scaling** method [15] is possible: For any magnetic source, the calculation can be carried out at a higher frequency f' (≤.5 MHz to guarantee the quasi-stationary character of the field). For this calculation, the electric conductivity $\sigma(f)$ of tissue must be taken into account for the frequency f (not f'). This calculation yields the electric field strength E' at the frequency f'. Now, by scaling the electric field strength due to

$$\vec{E}(\vec{r}) = f/f' \cdot \vec{E}'(\vec{r}) \tag{C.1}$$

the values for the frequency of interest (f) can be determined. Finally, the electric current density can be evaluated by applying Ohm's law:

$$J(\vec{r}) = \sigma(f) \cdot E(\vec{r}) \tag{C.2}$$

Validation procedure:

The situation depicted in Figure C.3 shall be considered. As body model, an homogeneous cuboid with edge length $d_x = d_y = 0{,}4$ m, $d_z = 1{,}8$ m and an electric conductivity $\sigma = 0{,}1$ S/m shall be investigated at a frequency $f = 50$ Hz.

NOTE The application of the frequency scaling method [15] is possible.

As field source, a square loop with a current $I = 1{,}0$ A and an edge length of 50 mm shall be considered with a distance of 10 mm in front of the cuboid (see Figure C.3).

With the software tool to be used for the test procedure, the electric current density induced in the body model shall be calculated.

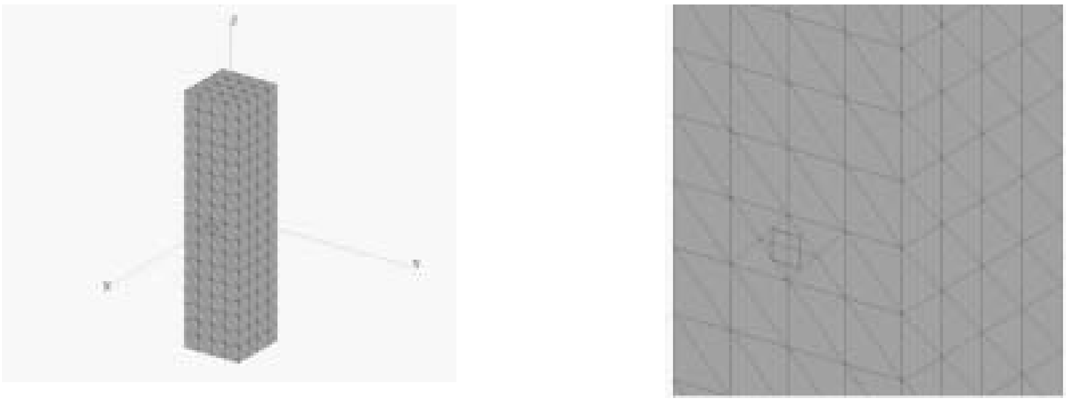


Figure C.3 – Test situation for validation – Current loop in front of a cuboid

The magnetic field source shall yield a maximum electric current density inside the tissue in the range of

J_{max} = 8,65 µA/m² ± 10 %

The factor of ± 10 % includes all deviations of the approaches in the different software packages used (e.g. the minimum distance to the surface of the cuboid, for which a field calculation is possible). More information can be found in [15] and [19].

References:

As references, the results determined by means of two different software packages are given:

FEKO (method of moments, [16]):

$$J_{\max} = 9{,}1\ \mu\text{A/m}^2,$$

EMPIRE (finite difference time domain method, [17]):

$$J_{\max} = 8{,}2\ \mu\text{A/m}^2.$$

Figure C.4 – Distribution of the electric current density J in the planes x = + 0,20 m (left) and y = 0,0 m (right)

Additionally, the distribution of the electric current density J is given for the planes x = + 0,2 m (Figure C.4, left) and y = 0,0 m (Figure C.4, right). The colour scaling used is a logarithmic one, normalised on each maximum value and performing a dynamic range of 30 dB.

NOTE The FIT-based software package EMPIRE [17] has been used.

C.6 Representation of measured magnetic field distributions by equivalent sources

Applying unique theorem of field theory and Huygens principle [12,13], a distribution of *fictive* (equivalent) sources (e.g. magnetic elementary dipoles) can be found on (or inside) the surface of a volume to represent the *real* sources inside. Figure C.5 shows a block diagram of the method proposed:

First, the magnetic flux density ***B*** (magnitude and phase) is measured on a surface (e.g. a cylinder) around the device under test (DUT) at the frequency of interest, e.g. by using the "3D-scan" set-up described in [14].

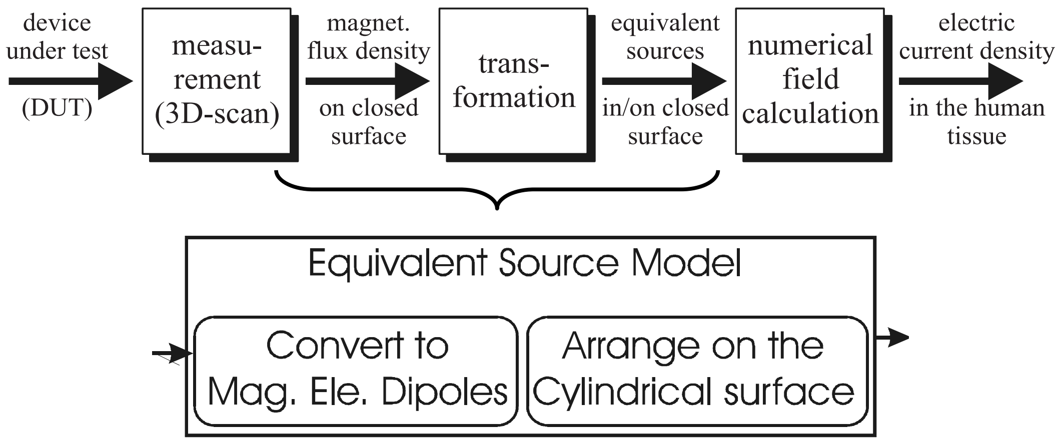


Figure C.5 – Block diagram of the method

In a second step, a numerical field transformation based on equation C.1 is carried out. The N magnetic elementary dipoles are e.g. located on the surface of the cylinder, on which the field data have been collected. Consequently, this yields the magnetic elementary dipole moment ***m*** instead of the measured ***B***.

$$\vec{H}_{res}(\vec{r}) = \sum_{j=0}^{N} \vec{H}_d(\vec{r},\vec{r}_{0,i}) \quad \text{with} \quad \vec{H}_d(\vec{r},\vec{r}_0) = -grad\left(\frac{\vec{m}\cdot(\vec{r}-\vec{r}_0)}{4\pi\mu_0\,|\vec{r}-\vec{r}_0|^3}\right) \tag{C.3}$$

Finally, the equivalent source model is used within a numerical field calculation to determine the induce electric current density inside the human body.

Now, from equation C.3, the unknown magnetic moments **m** can be determined with

$$\vec{H}_{res}(\vec{r}_j) = \vec{H}_{measured}(\vec{r}_j) \tag{C.4}$$

where $\vec{H}_{measured}(\vec{r}_j)$ denotes the measured magnetic field strength at the locations $\vec{r}_j$ on the surface considered. To solve the resulting system of linear equations, the conjugate gradient method [18] can be applied using a regularisation scheme.

The uncertainty caused by the transformation algorithm shall not exceed 5 %.

C.7 Induced current densities for different sizes of prolate spheroid

Here is an evaluation of induced current in three different size prolate spheroid solids: 60 cm by 30 cm, 120 cm by 60 cm and 160 cm by 80 cm, all full width and height dimensions. The modelling was performed using commercially available FEM Software.

The uniform field was simulated using coils that were large in relation to the prolate spheroids under consideration. Results are shown for both the generated magnetic field and the resultant induced current density, using 0,2 mS conductivity. The values are not specific to any one piece of equipment or any particular Guidelines. The ratio between the results shows how the modelling can be made using one size of spheroid and converted to another size using a multiplying factor.

C.7.1 Uniform magnetic field source

A very large set of Helmholtz coils, 5 m², were used to provide a uniform magnetic field at 58 kHz frequency. The degree of uniformity of the magnetic field was to within 1 % or less. Figure C.6 shows this geometry of the Helmholtz coils and prolate spheroid.

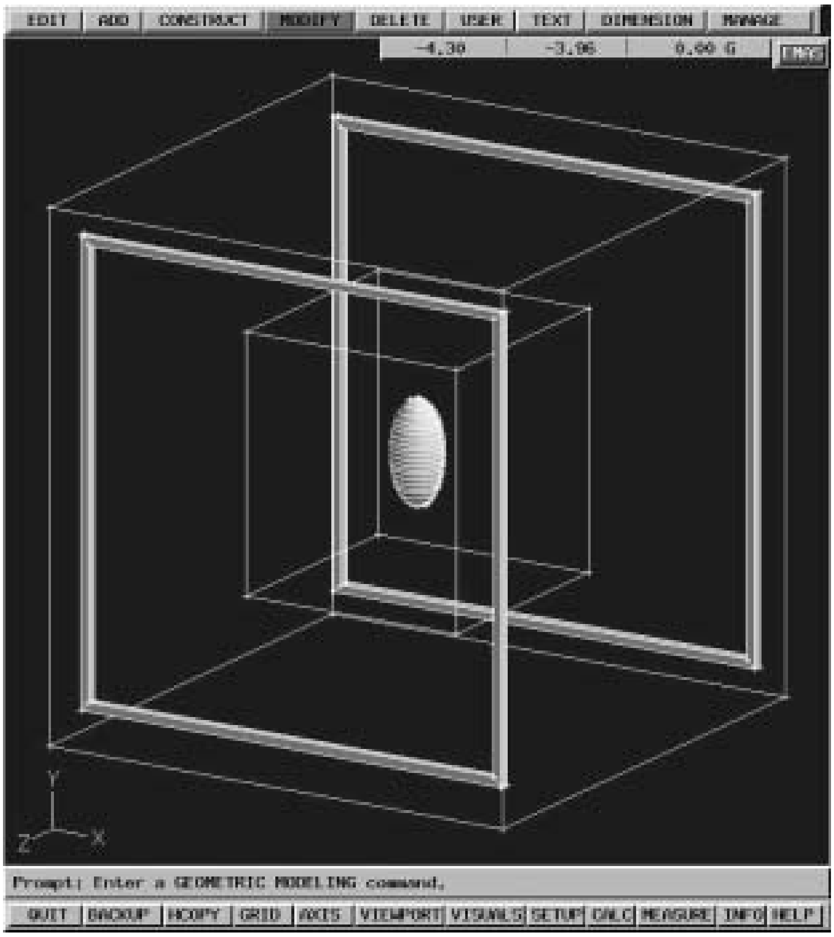


Figure C.6 – Helmholtz coils and prolate spheroid

C.7.2 Modelling results for a 60 cm by 30 cm Prolate Spheroid

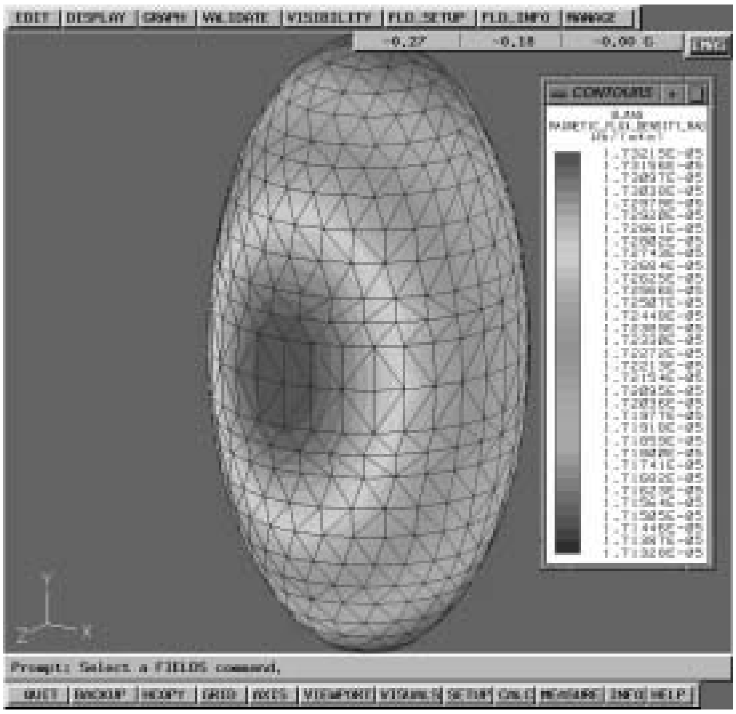


Figure C.7a – Magnetic field

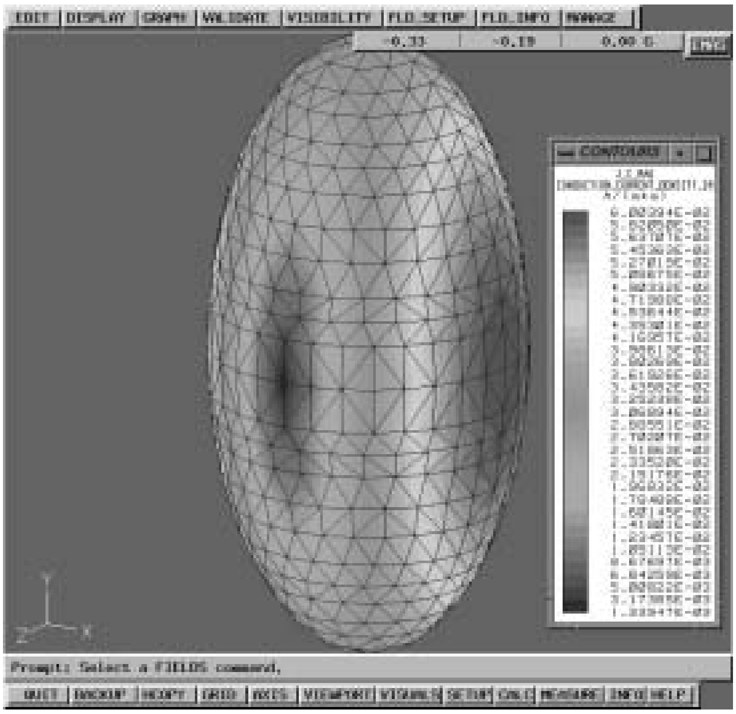


Figure C.7b – Induced current density

C.7.3 Modelling results for a 120 cm by 60 cm Prolate Spheroid

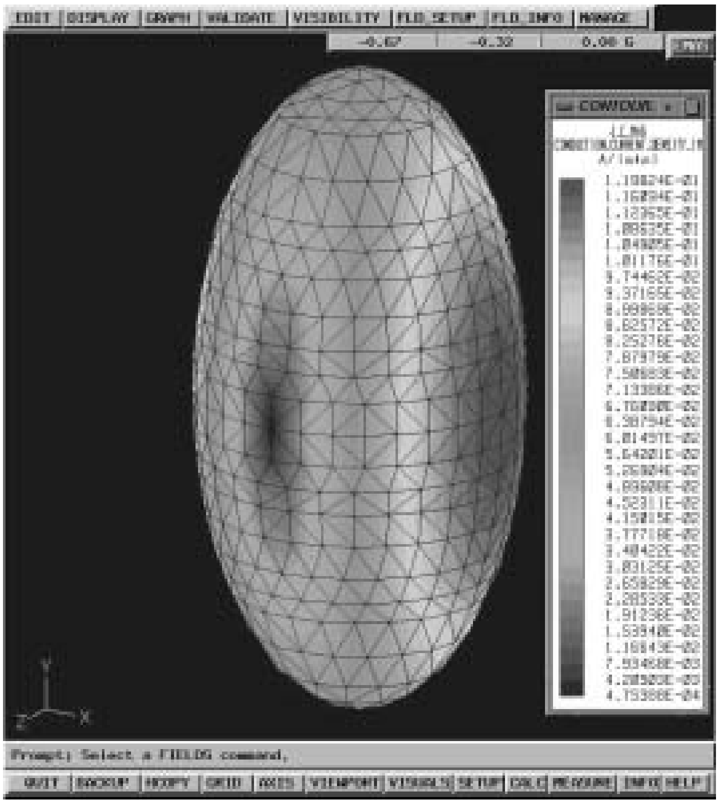


Figure C.8 – Induced current density

C.7.4 Modelling results for a 180 cm by 80 cm Prolate Spheroid

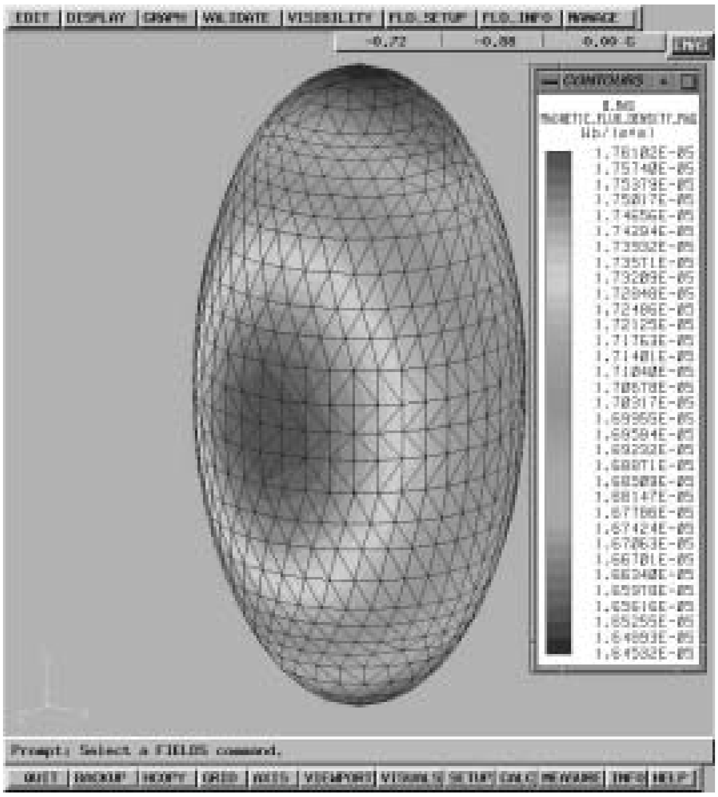


Figure C.9a – Magnetic field

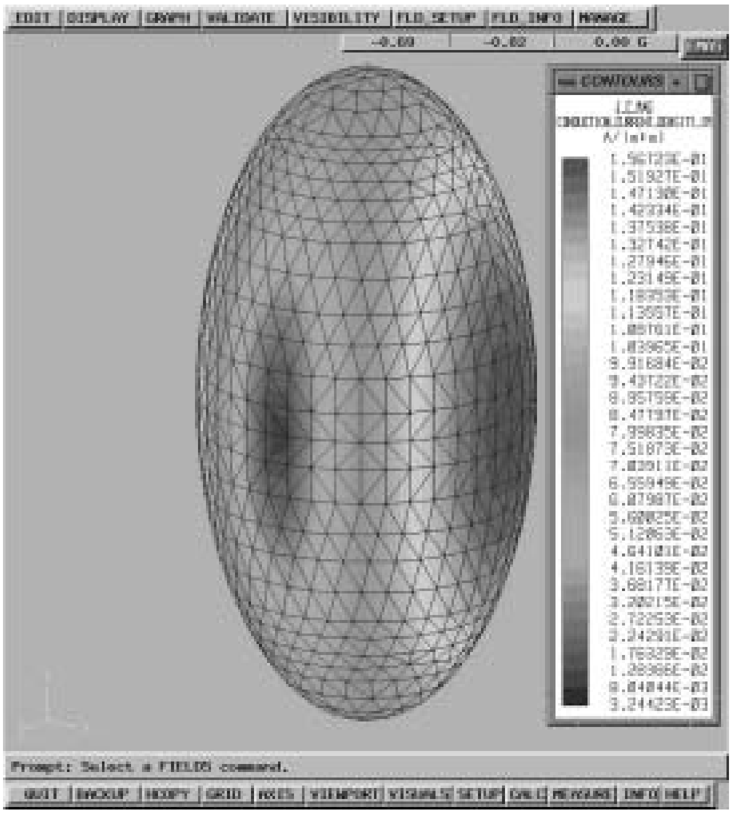


Figure C.9b – Induced current density

Table C.3 – Summary of results

Prolate spheroid size	Maximum magnetic field (used in the model)	Maximum induced current density	Ratio vs. 60 cm by 30 cm
60 cm by 30 cm	17,3 μT	60,0 mAm^{-2}	1,0
120 cm by 60 cm	17,5 μT	119,8 mAm^{-2}	2,0
160 cm by 80 cm	17,6 μT	156,7 mAm^{-2}	2,6

C.8 Induced current densities for the human body and hand (informative)

C.8.1 Uniform magnetic field

The following figures show the dimensions of the used numerical models for the homogenous human body and hand. In Figure C.10 a current loop is shown additionally as a possible source of non-uniform magnetic field.

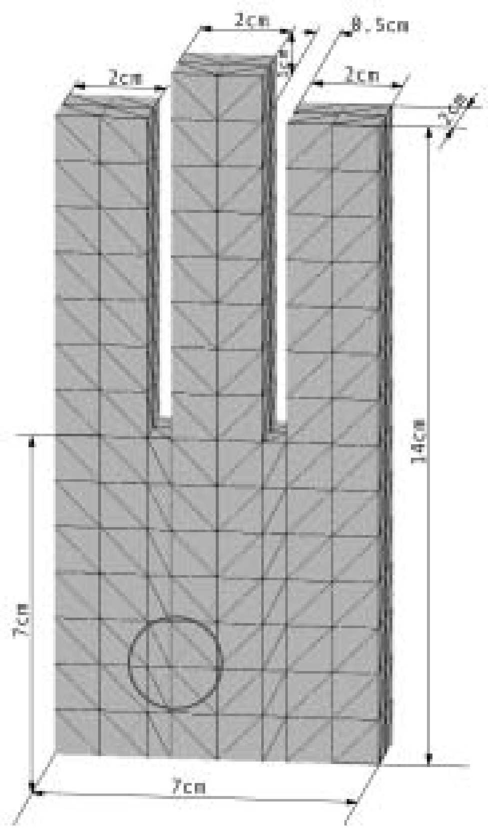


Figure C.10 – Numerical model of an homogenous hand

The ellipsoid (see Figure C.1) can be used to determine induced electric current densities numerically for a uniform magnetic field to compare it with numerical solutions.

EXAMPLE

A uniform magnetic field of $\hat{H}_{-x} = 112{,}5\ \frac{\text{A}}{\text{m}}$ at $f = 50$ Hz applied to a homogeneous body model (Figure C.2) with $\sigma = 0{,}37\ \frac{\text{S}}{\text{m}}$. The example is calculated using the method of moment [16].

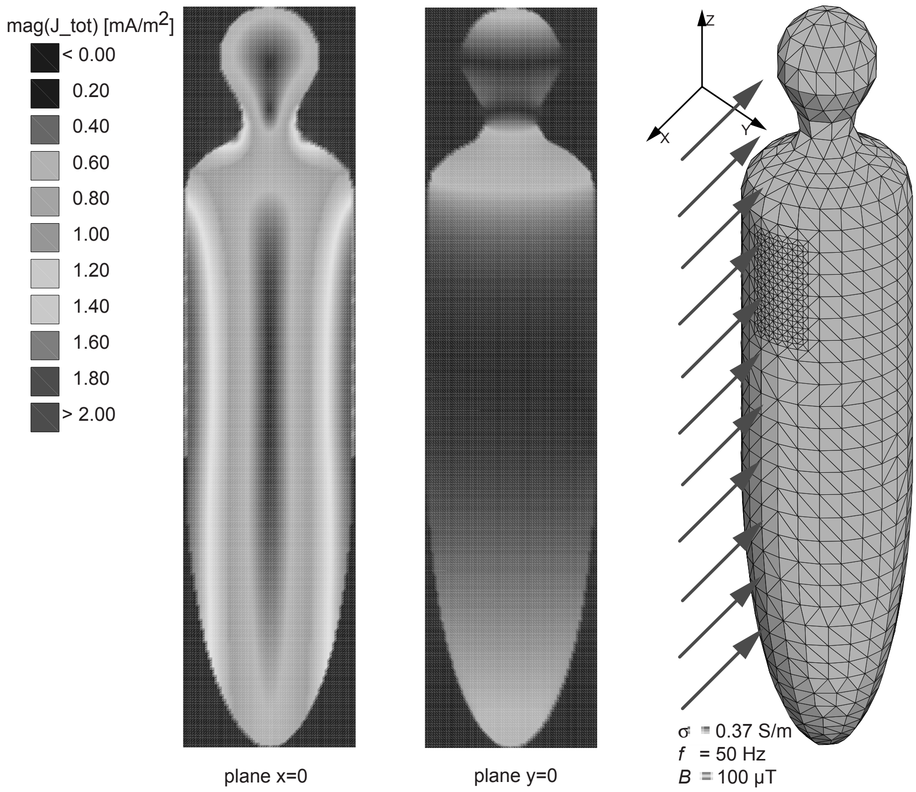


Figure C.11 – Distribution of induced electric current density

C.8.2 Non-uniform magnetic fields and calculation of the coupling factor *k*

The following list of sources for non-uniform magnetic fields cannot be complete but it gives an overview:

- circular current loops;
- rectangular current loops;
- single line currents;
- circular current coils;
- elementary dipoles.

However, only circular current loops were used as sources to calculate the coupling factors. Therefore the current loops of different diameter were positioned in a *worst-case* manner towards the numerical models. This is illustrated in Figure C.12.

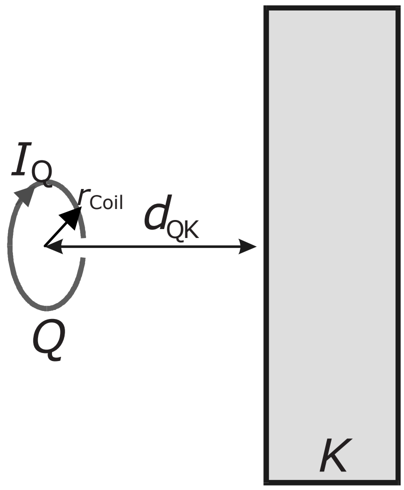


Figure C.12 – Position of source Q against model K

The coupling factor *k* gives the relation between the maximum induced electric current density $J_{max}(d_{QK})$ inside the numerical model and the maximum magnetic flux density measured at the same position of the model. The source current I_Q can be chosen arbitrarily but should be equal for the calculation of J_{max} and $B_{max,sensor}$. The evaluation of the coupling factor *k* depends therefore on the sensor used. For an arbitrary sensor area of A_{Sensor} the averaged magnetic flux density through it has to be calculated. The maximum $B_{max,\ Sensor}$ has to be chosen. Since the frequency *f* and the conductivity σ are linearly connected to the factor *k* it can be calculated as followed:

$$k(d_{QK}, f, \sigma) = \frac{J_{\max}(d_{QK}, f, \sigma)}{B_{\max,Sensor}(d_{QK}, A_{Sensor})}. \quad \text{(C.5)}$$

For the conductivity of the numerical homogenous hand model an average of σ = 0,1 S/m was evaluated.

For the conductivity of the homogenous body model σ = 0,2 S/m may be chosen. However, the non-uniformity of the fields and their very modest penetration into the body makes it possible to use σ = 0,1 S/m as well.

Other values for the conductivity σ can be achieved by [11].

For the determination of the coupling factor *k* in Annex C the method of moments (MoM) [16] as numerical technique was used.

EXAMPLE

For a circular coil with radius r_{Coil} = 20 mm in a distance *r* = 70 mm and a source current I_Q = 1 A one gets for the hand model (σ = 0,1 S/m and *f* = 50 Hz) the induced electric current density J_{max} = 262,5 µA/m². The averaged magnetic flux density for a 3 cm² sensor is calculated to $B_{max,sensor}$ = 3 cm² = 484,2 µT. The coupling factor *k* therefore calculates to

$$k(d_{QK} = 70 \text{ mm}, f = 50 \text{ Hz}, \sigma = 0{,}1 \tfrac{\text{S}}{\text{m}}) = \frac{262{,}5 \ \frac{\mu\text{A}}{\text{m}^2}}{484{,}2 \ \mu\text{T}} . \quad \text{(C.6)}$$

$$k(d_{QK} = 70 \text{ mm}, f = 50 \text{ Hz}, \sigma = 0{,}1 \tfrac{\text{S}}{\text{m}}) = 0{,}54212 \frac{\text{A/m}^2}{\text{T}}$$

C.9 References

[1] "Induced Current Densities from Low-Frequency Magnetic Fields in a 2 mm Resolution, Anatomically Realistic Model of the Body", P. J. Dimbylow, Phys. Med. Biol., Vol. 43, pp. 221-230, 1998.

[2] "Inter-Laboratory Comparison of Numerical Dosimetry for Human Exposure to 60 Hz Electric and Magnetic Fields", M. A. Stuchly and O. P. Gandhi, Publication data to be advised

[3] ICRP 66 (1994), International Commission for Radiological Protection, 1994
"The Specific Resistance of Biological Material – A Compendium of Data for the Biomedical Engineer", L. A. Geddes and L. E. Baker, Medical and Biological Engineering, Vol. 5 pp 271-293, 1967

[4] "Dielectric Properties of Tissues", K. R. Foster and H. P. Schwan, Handbook of Biological Effects of Electromagnetic Fields, Second Edition, Ed. E. Polk and E. Postow, CRC Press, Boca Raton, Fla., USA, 1995

[5] Radio requency Radiation Dosimetry Handbook", 4th Ed., C. H. Durney, H. Massoudi, and M. F. Iskander, USAF/SAM, Brooks AFB, TX, USA, 1986

[6] Electric Properties of Biological Substances – Tabulated", M. A. Stuchly and S. S. Stuchly, Journal of Microwave Power, Volume 15(1), pp 19-26, 1980

[7] The Dielectric Properties of Biological Tissues: I. Literature survey", C. Gabriel, S. Gabriel, and E. Courthout, Phys. Med. Biol. 41 (11), pp 2231-2250, 1996

[8] The Dielectric Properties of Biological Tissues: II. Measurement in the frequency range 10 Hz to 20 GHz", S. Gabriel, R. W. Lau, and C. Gabriel, Phys. Med. Biol. 41 (11), pp 2251-2269, 1996

[9] The Dielectric Properties of Biological Tissues: III. Parametric models for the dielectric spectrum of tissues", S. Gabriel, R. W. Lau, and C. Gabriel, Phys. Med. Biol. 41 (11), pp 2271-2293, 1996

[10] Compilation of the Dielectric Properties of Body Tissues at RF and Microwave Frequencies", C. Gabriel and S. Gabriel, Report AL/OE-TR-1996-0037, Armstrong Laboratory, Radiofrequency Radiation Division, Brooks Air Force Base, Texas 78235, USA, 1996; (Internet Site: http://www.brooks.af.mil; http://www.brooks.af.mil/AFRL/HED/hedr/reports/dielectric/Title/Title.html)

[11] The Electromagnetic Wave Research Institute of the Italian National Research Council, Via Panciatichi 64, 50127 Florence, Italy (Internet Site: http://www.iroe.fi.cnr.it ; http://sparc10.iroe.fi.cnr.it/tisspro)

[12] H-O. Ruoss, W. Spreitzer, S. Nishizawa, S. Messy and M. Klar, "Efficient determination of current densities induced in the human body from measured low-frequency inhomogeneous magnetic fields", Microwave and Optical Technology Letters, vol. 29, no. 4, pp. 211-213, May 20, 2001

[13] Spreitzer, S. Nishizawa, H.-O. Ruoss, S. Messy and F. Landstorfer, "Equivalent source model for household appliance emitted low frequency magnetic field", IEEE Transaction on MTT, 2000 (Submitted)

[14] U. Kampet and W. Hiller, "Measurement of magnetic flux densities in the space around household appliances", in: Proceedings of NIR 99, Nichtionisierende Strahlung, 31. Jahrestagung des Fachverbandes für Strahlenschutz, Köln, vol. II, pp. 885-891, 1999

[15] C.M. Furse and O.P. Gandhi, "Calculation of electric fields and currents induced in a millimeter-resolution human model at 60 Hz using the FDTD method", Bioelectromagnetics, vol. 19, pp. 293-299, 1998

[16] U. Jakobus, "Erweiterte Momentenmethode zur Behandlung kompliziert aufgebauter und elektrisch grosser elektromagnetischer Steuprobleme", Fortschrittsberichte VDI, Reihe 21, Nr.171, VDI Verlag, Duesseldorf, 1995

[17] Programm EMPIRE, http://www.imst.de/

[18] J.R. Shewchuck, "An introduction to the conjugate gradient method without the agonizing pain", School of Computer Science, Carnegie Mellon University, Pittsburgh, 1994

[19] H.-O. Ruoß and U. Kampet "Numerical calculation of current densities induced in the human body caused by low frequency inhomogeneous magnetic sources", Kleinheubacher Berichte 2001, Band 144, pp. 155-162, 2001

Annex D
(informative)

Measurements of physical properties and body currents

D.1 Measurement of body current

Measurement devices for body current may be carried out in two categories:

- measurement devices for body to ground current;
- measurement devices for contact current.

D.2 Measurement of induced body currents

Body currents are induced currents resulting from exposure of the body to RF fields in the absence of contact with objects other than the ground. The two principal techniques used for measuring body currents include clamp-on type (solenoidal) current transformers for measuring current flowing in the limbs, and parallel plate systems that permit the measurement of currents flowing to ground through the feet.

Clamp-on current transformer instruments have been developed that can be worn. The meter unit is mounted either directly on the transformer or connected through a fibre-optic link to provide a display of the current flowing in a limb around which the current transformer is clamped. Current sensing in these units may be accomplished using either narrowband techniques, e.g., spectrum analysers or tuned receivers which offer the advantage of being able to determine the frequency distribution of the induced current in multi-source environments, or broadband techniques using diode detection or thermal conversion.

Instruments have been designed to provide true RMS indications in the presence of multiple frequencies and/or amplitude-modulated waveforms.

The upper frequency response of current transformers is usually limited to about 100 MHz however air cored transformers (as opposed to ferrite-cored), have been used to extend the upper frequency response of these instruments. Whilst air-cored transformers are lighter and therefore useful for longer-term measurements, they are significantly less sensitive than ferrite-cored devices.

An alternative to the clamp-on device is the parallel plate system. In this instrument, the body current flows through the feet to a conductive top plate, through some form of current sensor mounted between the plates, and thereby to ground. The current flowing between the top and bottom plates may be determined by measuring the RF voltage drop across a low impedance resistor. Alternatively, a small aperture RF current transformer or a vacuum thermocouple may be used to measure the current flowing through the conductor between the two plates.

Instruments with a flat frequency response between 3 kHz and 100 MHz are available. There are several issues that should be considered when selecting an instrument for measuring induced current.

Firstly, stand-on meters are subject to the influence of electric-field induced displacement currents from fields terminating on the top plate. Investigations have shown that apparent errors arising in the absence of a person are not material to the operation of the meters when a person is present.

Secondly, the sum of both ankle currents measured with clamp-on type meters tends to be slightly greater than the corresponding value indicated with plate type meters. The magnitude of this effect, which is a function the RF frequency and meter geometry, is not likely to be material. Nonetheless, the more accurate method of assessing limb currents is the current transformer. The precise method of measurement may depend upon the requirements of protection guidelines against which compliance assessments are made.

Thirdly, the ability to measure induced currents in limbs under realistic grounding conditions such as found in practice need to be considered. In particular, the differing degree of electrical contact between the ground and bottom plate of the parallel plate system and the actual ground surface may affect the apparent current flowing to ground.

Measurements can be made using antennas designed to be equivalent to a person. This enables a standardised approach to be used and permit current measurements to be made without the need for people to be exposed to potentially hazardous currents and fields.

D.3 Measurement of contact current

The current measurement device has to be inserted between the hand of the person and the conductive object. The measurement technique may consist of a metallic probe (definite contact area) to be held by hand at one end of the probe while the other end is touched to the conductive object. A clamp-on current sensor (current transformer) as described in D.2 can be used to measure the contact current, which is flowing into the hand in contact with the conductive object.

In the case where excessively high currents are expected an electrical network of resistors and capacitors can simulate the body's equivalent impedance.

Alternative methods are:

- measurement of the potential difference (voltage drop) across a non-inductive resistor (resistance range of 5 Ω - 10 Ω) connected in series between the object and the metallic probe holding in hand;
- a thermocouple millimetre placed directly in series.

The wiring connections and the current meter must be set up in such way that interference and errors due to "pick-up" are minimised.

D.4 Measurement of touch voltage

The touch voltage (no-load-voltage) is measured by means of a suitable voltmeter or oscilloscope for the frequency range under consideration. The measurement devices are connected between the conductive object charged by field induced voltage and reference potential (ground). The input impedance of the voltmeter must not be smaller than 10 kΩ.

D.5 References

– “Induced Current Densities from Low-Frequency Magnetic Fields in a 2 mm Resolution, Anatomically Realistic Model of the Body”, P. J. Dimbylow, *Phys. Med. Biol.,* Vol. 43, pp. 221-230, 1998

Comparison of Numerical Dosimetry for Human Exposure to 60 Hz Electric and Magnetic Fields”, M. A. Stuchly and O. P. Gandhi, *Publication data to be advised*

– ICRP 66 (1994), *International Commission for Radiological Protection,* 1994

– “The Specific Resistance of Biological Material – A Compendium of Data for the Biomedical Engineer”, L. A. Geddes and L. E. Baker, *Medical and Biological Engineering*, Vol. 5 pp 271-293, 1967

– “Dielectric Properties of Tissues”, K. R. Foster and H. P. Schwan*, Handbook of Biological Effects of Electromagnetic Fields,* Second Edition, Ed. E. Polk and E. Postow, CRC Press, Boca Raton, Fla., USA, 1995

– “Radio Frequency Radiation Dosimetry Handbook”, 4th Ed., C. H. Durney, H. Massoudi, and M. F. Iskander, *USAF/SAM, Brooks AFB, TX*, *USA,* 1986

– “Dielectric Properties of Biological Substances – Tabulated”, M. A. Stuchly and S. S. Stuchly, Journal of Microwave Power, Volume 15(1), pp 19-26, 1980

– “The Dielectric Properties of Biological Tissues: I. Literature survey”, C. Gabriel, S. Gabriel, and E. Courthout, *Phys. Med. Biol*. 41 (11), pp 2231-2250, 1996

– “The Dielectric Properties of Biological Tissues: II. Measurement in the frequency range 10 Hz to 20 GHz”, S. Gabriel, R. W. Lau, and C. Gabriel, *Phys. Med. Biol.* 41 (11), pp 2251-2269, 1996

– “The Dielectric Properties of Biological Tissues: III. Parametric models for the dielectric spectrum of tissues”, S. Gabriel, R. W. Lau, and C. Gabriel, *Phys. Med. Biol*. 41 (11), pp 2271-2293, 1996

– “Compilation of the Dielectric Properties of Body Tissues at RF and Microwave Frequencies”, C. Gabriel and S. Gabriel, *Report AL/OE-TR-1996-0037,* Armstrong Laboratory, Radiofrequency Radiation Division, Brooks Air Force Base, Texas 78235, USA, 1996; (*Internet Site: http://www.brooks.af.mil; http://www.brooks.af.mil/AFRL/HED/hedr/reports/dielectric/Title/Title.html)*

– The Electromagnetic Wave Research Institute of the Italian National Research Council, Via Panciatichi 64, 50127 Florence, Italy (*Internet Site: http://www.iroe.fi.cnr.it ; http://sparc10.iroe.fi.cnr.it/tissprop)*

Annex E
(informative)

SAR

E.1 Specific Absorption Rate (SAR) measurement procedures

E.1.1 Electric field measurement procedures

The SAR is also proportional to the squared RMS electric field strength E (V/m) inside the exposed tissue:

$$SAR = \sigma E^2 / \rho$$

where σ (S/m) is the conductivity and ρ (kg/m³) is the mass density of the tissue material at the position of interest. Using an isotropic electric field probe, the local SAR inside an irradiated body model can be determined. By moving the probe and repeating the electric field measurements in the whole body or in a part of the body, the SAR distribution and the whole-body or partial-body averaged SAR values can be determined. A single electric field measurement takes only a few seconds, which means that three-dimensional SAR distributions can be determined with high spatial resolution and with a reasonable measurement time (typically less than an hour). In Annex B, procedures for evaluations of local peak SAR for handheld radio transmitters and radio base stations are defined.

E.1.2 Temperature measurement procedures

The SAR is proportional to initial rate of temperature rise dT/dt (°C/s) in the tissue of an exposed object:

$$SAR = c\Delta T/\Delta t$$

where c is the specific heat capacity of the tissue material (J/kgC). Using certain temperature probes, the local SAR inside an irradiated body model can be determined. One or more probes are used to determine the temperature rise ΔT during a short exposure time Δt (typically less than 30 s to prevent heat transfer). The initial rate of temperature rise is approximated by ΔT/Δt, and the local SAR value is calculated for each measurement position. By repeating the temperature measurements in the whole body or in a part of the body, the SAR distribution and the whole-body or partial-body averaged SAR values can be determined.

Three-dimensional SAR-distribution measurements are very time consuming due to the large number of measurement points. To achieve a reasonable measurement time the number of points has to be limited. This means that it is very difficult to measure strongly non-uniform SAR distributions accurately. The accuracy of temperature measurements may also be affected by thermal conduction and convection during measurements, or between measurements.

E.1.3 Calorimetric measurement procedures

The whole-body average SAR can be determined using calorimetric methods. In a normal calorimetric measurement, a full-size or scaled body model at thermal equilibrium is irradiated for a period of time. A calorimeter is then used to measure the heat flow from the body, until the model is at thermal equilibrium again. The obtained total absorbed energy is then divided by the exposure time and the mass of the body model, which gives the whole-body SAR. The calorimetric twin-well technique uses two calorimeters and two identical body models. One of the models is irradiated, and the other one is used as a thermal reference. This means that the measurement can be performed under less well-controlled thermal conditions than a normal calorimetric measurement.

Calorimetric measurements give rather accurate determinations of whole-body SAR, but do not give any information about the internal SAR distribution. To get accurate results a sufficient amount of energy deposition is required. The total time of a measurement, which is determined by the time to reach thermal equilibrium after exposure, may be up to several hours. Partial-body SAR can be measured by using partial-body phantoms and small calorimeters.

Annex F
(informative)

Measurement of E & H fields

F.1 Measurement of external electromagnetic fields

F.1.1 General considerations

The measurement of external fields with regard to human exposure assessment will depend upon the objective. In the first instance it may be that the measurements are simply to assess compliance with external field strength reference level values contained in exposure guidelines. For some guidelines additional information may be required to enable calculation of the spatial averaging of inhomogeneous field distributions. In other cases detailed field distribution data may be needed to provide input to other analytical or computational techniques for assessing compliance with the basic quantities underpinning particular guidelines. The approaches used and the spatial resolution of instrumentation used to carry out these tasks may differ substantively.

Prior to making measurements, one should estimate the expected field strength and determine the type of instrument required. Additional approaches and equations for calculating field strength in various situations are given in Annex A. The measurement procedures to be used may differ, depending on the source and propagation information available.

If the information is adequate, then the surveyor, after making estimates of expected field strengths and selecting an instrument, may proceed with the survey. The surveyor should use a high-power probe with the range switch set on the most sensitive scale. The high-intensity field areas, e.g., the main beam of a directional antenna, should be approached from a distance to avoid probe burnout. The surveyor then gradually proceeds to move progressively closer to the regions of higher field strength. Extreme care should be exercised to avoid overexposure of the surveyor and survey instrument.

On the other hand, if the information is not well defined (for example, reports of strong, intermittent interference), then it may be difficult to make a hazard survey without first conducting an empirical hazard assessment. A survey for potentially hazardous fields of unknown frequency, modulation, distribution within an area, etc. may require use of several instruments.

When performing a measurement, the overall measurement uncertainty shall be predicted and evaluated. All possible sources of uncertainty including the instrumentation specifications and the specific situation parameters shall be taken into account.

F.1.2 Equivalent field strength

F.1.2.1 Electric field strength

The electric component of the electromagnetic field can be easily measured using suitable antennas, e.g. bi-conical, log-periodic etc. However, for exposure assessment, small elementary dipoles are generally used as sensors in order to minimally perturb the field and to ensure a good spatial resolution. Directional probes contain only one dipole, whereas isotropic probes contain three orthogonal dipoles.

If a single dipole is used, three measurements should be performed in three orthogonal directions to obtain the different components of the field. The total E-field would be given by the following formula:

$$E = \sqrt{E_x^2 + E_y^2 + E_z^2}$$

F.1.2.2 Magnetic field strength

The magnetic component of the electromagnetic field is usually measured with loop sensors, as the current induced in the loop is proportional to the magnetic field strength crossing the loop. Here again, for exposure assessment, small loops are generally used as sensors in order to disturb as little as possible the field and to ensure a good spatial resolution. Directional probes (one loop) are widely used, but many isotropic probes exist with three orthogonal loops.

If a single loop is used, three measurements should be performed in three orthogonal planes to obtain the different components of the field. The total H-field would be given by the following formula:

$$H = \sqrt{H_x^2 + H_y^2 + H_z^2}$$

F.1.2.3 Broadband measurements

If several frequencies (and varying modulations) are present in the frequency range to be observed, and if the derived levels are the same for each of these frequencies, then either the peak values or the RMS values (irrespective of signal shape) can be measured directly with appropriate broadband measuring equipment. However it must be ensured that the bandwidth of the instrument is sufficiently wide to record all the relevant frequencies. Furthermore, the selected observation period must be long enough to record the maximum peak/RMS values that are generated as a result of interaction of the superimposed frequency components.

For practical reasons, most of the measurement instruments on the market are broadband systems, in which case the above-mentioned precautions have to be taken to ensure accurate results.

F.1.2.4 Narrowband measurements

If several Frequencies (and varying modulations) are present in the frequency range to be observed, and if the derived levels are the same for each of these frequencies, then the peak values and/or the RMS values at each frequency can be measured directly with frequency-selective measurement equipment. In this case, it should be noted that peak values for the individual (independent from each other) frequencies should be added linearly, in order to determine the total peak value, whereas RMS values for the individual frequencies should be added geometrically, in order to determine the total RMS value.

NOTE Measurement of peak value is not recommended because the limits are expressed in rms and because of problems with reproducibility.

If the derived levels are not the same for all the frequency components to be evaluated, then a sufficiently narrow bandwidth should be chosen, to ensure that the influence of the change in the derived value within the frequency range covered by the instrument is negligible.

If measurements are carried out in the time domain (using a transient recorder) and the frequency spectrum is calculated by Fourier transformation, then an adequate frequency resolution must be ensured in order to facilitate the evaluation of limiting values (this is not applicable when frequencies are independent of each other).

Annex G
(informative)

Source modelling

G.1 Numerical modelling

G.1.1 Description of available methods

Analytical procedures can only be used to calculate the electromagnetic properties for a few special cases and geometries. To solve general problems, numerical techniques have to be applied. The most common numerical procedures to calculate the electromagnetic fields from a transmitting source or the internal fields and the specific absorption rate in biological bodies, are listed below. A brief description is also given for some of these methods. Which of the numerical techniques that is most appropriate for a certain problem, depends on the frequency range considered, the geometrical structures to be modelled, and the type exposure situation (near-field or far-field). References [1-7] contain further information about these techniques and their application.

Numerical modelling methods:

- physical optics (PO);
- physical theory of diffraction (PTD);
- geometrical optics (GO);
- geometrical theory of diffraction (GTD);
- uniform theory of diffraction (UTD);
- method of equivalent currents (MEC);
- method of moments (MoM);
- multiple multipole method (MMP);
- finite-difference time-domain method (FDTD);
- finite element method (FEM);
- impedance method.

G.1.1.1 Method of Moments (MoM)

The method of moments is a technique, which has been extensively used to solve electromagnetic problems and to make SAR calculations in block models of biological bodies. In MoM, the electric fields inside a biological body are calculated by means of a Green's function solution of Maxwell's integral equations.

G.1.1.2 Fast Fourier Transform/Conjugate Gradient method (FFT/CG)

The FFT/CG method is a further development of the method of moments. Iterative algorithms based on FFT and the gradient procedure is used to solve linear equations derived from the method of moments.

G.1.1.3 Finite-Difference Time-Domain method (FDTD)

FDTD is a numerical method to solve Maxwell's differential curl equations in the time domain. It can be used to calculate internal and external electromagnetic fields and SAR distribution in biological bodies for both near-field or far-field exposures. In FDTD, both time and space are discretised, and a biological body is modelled by assigning the permitivity and conductivity values to the space cells it occupies. The computer memory required is proportional to the number of space cells. FDTD is considered the most promising SAR calculation method, but for accurate calculations very powerful computers are needed.

G.1.1.4 Multiple multipole method (MMP)

MMP is based on analytical solutions to field equations, which have a multipole at one point in space, and is used in conjunction with the generalized multipole technique (GMP). The MMP procedure is especially suitable for the simulation of so-called "lossy scattering" bodies, which are near to radiation sources, i.e. within the immediate near-field.

G.1.1.5 Impedance method

The impedance method has been successfully used to solve dosimetric problems where quasi-static approximations can be made. For calculations of SAR in human bodies, this method has proven to be very effective at frequencies up to 40 MHz. In the impedance method, the biological body is modelled by a three-dimensional network of complex impedances.

G.2 Field strength calculations

Most of the methods listed above can be used to calculate field strength levels from electromagnetic radiators. The accuracy of the results depends very much on how well the radiator (for example antenna) is modelled. If objects near the radiator, between the radiator and the prediction point, or close to the point of field strength prediction may affect the field strength levels significantly, such objects should also be modelled.

G.3 Specific absorption rate calculations

Due to the difficulty to measure the whole-body averaged or local peak SAR in many exposure situations, numerical calculations several of the numerical techniques mentioned above can be used for estimation of the specific absorption rate distribution in a biological body exposed to either near-field or far-field electromagnetic radiation, for example the Finite-difference time-domain method (FDTD), Method of moments, and the Multiple multipole method (MMP).

Which of these methods that is most appropriate for a particular problem, depends e.g. on the frequency, the exposure conditions, the size of the exposed object, the required accuracy, and the maximum tolerable calculation time. Each method requires experience in biophysics and numerical analysis.

To use any of these models, a three-dimensional geometric numerical model of the exposed body, or part of the body, is required. The electrical properties at the exposure frequency should be known for the different parts of the body. Depending on the required accuracy, models with different complexity may be used. In some situations, simple shapes like spheres and cylinders are appropriate to model the body. The dielectric properties of human tissues are given in the literature [8]. Using magnetic resonance (MR) images of a human body, very complex and accurate numerical body models can be developed. MR models with several different tissue types and a spatial resolution of less than a few millimetres have been used for FDTD calculations of the SAR distribution in humans exposed to electromagnetic fields from handheld radio transmitters [9,10].

Bibliography

EN 50361	Basic standard for the measurement of Specific Absorption Rate related to human exposure to electromagnetic fields from mobile phones (300 MHz - 3 GHz)
IEC 61786	Measurement of low-frequency magnetic and electric fields with regard to exposure of human beings – Special requirements for instruments and guidance for measurement
IEC 62334 [2)]	Measurement and assessment of human exposure to high frequency (9 kHz to 300 GHz) electromagnetic fields

2) Under consideration.

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